



Dissolved and suspended organic matter dynamics in the Cape Verde Frontal Zone (NW Africa)

S. Valiente^{a,b,*}, B. Fernández-Castro^{b,c}, R. Campanero^{a,b}, A. Marrero-Díaz^d, A. Rodríguez-Santana^d, M.D. Gelado-Cabellero^e, M. Nieto-Cid^{b,f}, A. Delgado-Huertas^a, J. Arístegui^g, X. A. Álvarez-Salgado^b

^a CSIC Instituto Andaluz de Ciencias de la Tierra (IACT-CSIC), Granada, Spain

^b CSIC Instituto de Investigaciones Mariñas (IIM-CSIC), Vigo, Spain

^c Ocean and Earth Science, National Oceanography Center Southampton, University of Southampton, Southampton, UK

^d Departamento de Física, Universidad de Las Palmas de Gran Canaria, Las Palmas de Gran Canaria, Spain

^e Departamento de Química, Universidad de Las Palmas de Gran Canaria, Las Palmas, Spain

^f Instituto Español de Oceanografía (IEO), Centro Oceanográfico de A Coruña, A Coruña, Spain

^g Instituto de Oceanografía y Cambio Global, Universidad de Las Palmas de Gran Canaria, Telde, Spain

ARTICLE INFO

Keywords:

Dissolved organic matter
Particulate organic matter
Water masses
Carbon cycling
Nitrogen cycling
Cape Verde Frontal Zone

ABSTRACT

The Cape Verde Frontal Zone (CVFZ) is a highly dynamic region located in the southern boundary of the Canary Current Eastern Boundary Upwelling Ecosystem. Due to the interaction of the Cape Verde Front with the Mauritanian coastal upwelling, the area features large vertical and horizontal export fluxes of organic matter. While the flux, composition and biogeochemical role of sinking organic matter have been thoroughly studied, much less attention has been paid to the dissolved (DOM) and suspended particulate (POM) organic matter fractions. Full-depth profiles of DOM and POM were recorded during an oceanographic cruise in the CVFZ, with four consecutive transects defining a box embracing the giant filament of Cape Blanc and the Cape Verde front. The distributions of DOM and POM and their C:N stoichiometric ratios in the epipelagic layer were strongly influenced by the position of the transects relative to the giant filament and the front. Geographical heterogeneity in POM and DOM distributions and elemental composition was also observed within each of the different water masses of contrasting origin present in the area (North and South Atlantic Central Water, Subpolar Mode Water, Mediterranean Water, Antarctic Intermediate Water, Labrador Sea Water and North East Atlantic Deep Water). These facts suggest that water masses properties are re-shaped by biogeochemical processes occurring within the CVFZ. Nevertheless, our analysis indicates that DOM and POM mineralisation represents only 8.1% of the inorganic carbon and 17.8% of the inorganic nitrogen produced by the local mineralisation of organic matter. Intense lateral export of POM and DOM out of the boundaries of the CVFZ is the likely reason behind these low contributions, which confirm the prominent role of sinking fluxes of organic matter for mineralisation processes in this region. The DOM distribution in the CVFZ interior is apparently affected by the dissolution of fast sinking particles.

1. Introduction

The Eastern Boundary Upwelling Ecosystem (EBUE) associated with the Canary Current extends from the northern tip of the Iberian Peninsula at 43°N to South of Senegal at about 10°N (Arístegui et al., 2009). The intensity and persistence of upwelling-favourable winds increase with decreasing latitude, evolving from weak and seasonal to the North to strong and year-round to the South. The Canary Current EBUE is also characterized by an intense mesoscale activity in the form of upwelling

filaments associated with prominent capes (Álvarez-Salgado et al., 2007; Lovecchio et al., 2018; Santana-Falcón et al., 2020) and island eddies (Barton et al., 1998; Sangrà et al., 2009; Cardoso et al., 2020). These mesoscale structures may export large amounts of organic matter produced over the shelf several hundreds of kilometres offshore (Gabric et al., 1993; Ohde et al., 2015; Lovecchio et al., 2018). Mesoscale activity is particularly intense in the southern boundary of the Canary Current EBUE due to the confluence of the Mauritanian upwelling and the Cape Verde Front (CVF). The CVF separates the eastern margins of the subtropical and tropical Atlantic gyres (Pelegri and Peña-Izquierdo,

* Corresponding author.

<https://doi.org/10.1016/j.pocean.2021.102727>

Received 20 April 2021; Received in revised form 21 August 2021; Accepted 3 December 2021

Available online 8 December 2021

0079-6611/© 2021 Elsevier Ltd. All rights reserved.

List of acronyms

AA	Modified Antarctic Intermediate Water
AOU	Apparent oxygen utilization
Chl-a	Chlorophyll <i>a</i>
CVF	Cape Verde Front (when referring exclusively to the Front)
CVFZ	Cape Verde Frontal Zone (when referring to the entire study area)
DCM	Deep chlorophyll maximum
DOC	Dissolved organic carbon
DOM	Dissolved organic matter
DON	Dissolved organic nitrogen
EBUE	Eastern Boundary Upwelling Ecosystem
ENACW	Eastern North Atlantic Central Water
ENACW_15	Eastern North Atlantic Central Water of 15 °C
ENACW_12	Eastern North Atlantic Central Water of 12 °C
LNEADW	Lower North East Atlantic Deep Water
LSW	Labrador Sea Water

MMW	Madeira Mode Water
MW	Mediterranean Water
NEADW	North East Atlantic Deep Water
NO	Conservative chemical parameter, obtained as $NO = O_2 + R_N \cdot NO_3^-$
OMP	Optimum multiparameter analysis
POC	Particulate organic carbon
POM	Particulate organic matter
PON	Particulate organic nitrogen
R_N	Stoichiometric conversion factor of nitrate to dissolved oxygen, $9.3 \text{ mol } O_2 \text{ mol } NO_3^{-1}$
SACW	South Atlantic Central Water
SACW_12	South Atlantic Central Water of 12 °C
SACW_18	South Atlantic Central Water of 18 °C
SE	Standard error
SPMW	Subpolar Mode Water
TDN	Total dissolved nitrogen
UNEADW	Upper North East Atlantic Deep Water

2015) and favours the horizontal transport from the coast, contributing to developing the giant filament of Cape Blanc (21°N). This structure exports offshore massive amounts of biogenic organic matter produced in the Mauritanian coast (Van Camp et al., 1991; Gabric et al., 1993). At the CVF, Eastern North Atlantic Central Water (ENACW) encounters the slightly cooler, less saline, nutrient-rich and oxygen-poorer South Atlantic Central Water (SACW) (Zenk et al., 1991). In the frontal region, intense interleaving between the two water masses occurs (Pérez-Rodríguez et al., 2001; Martínez-Marrero et al., 2008), as well as the recurrent formation of meanders and eddies (Meunier et al., 2012; Alpers et al., 2013).

Upwelling filaments, meandering fronts and eddies affect the distribution and net community production of surface plankton communities in the Cape Verde Frontal Zone (CVFZ) (Olivar et al., 2016; Tiedemann et al., 2018; Arístegui et al., 2020). The organic matter transported by these structures is dominated by small suspended or slow sinking particulate organic matter, hereinafter suspended POM, and dissolved organic matter (DOM), while fast sinking POM (generally > 100 µm) is less susceptible to horizontal advection (Álvarez-Salgado and Arístegui, 2015). Differentiation between suspended and sinking POM is of fundamental importance because of the contrasting mechanisms involved in their dispersion (horizontal export for suspended POM vs. vertical export for sinking POM) and fate during offshore transport from the coast to the ocean interior (Helmke et al., 2005; Fischer et al., 2009; Álvarez-Salgado and Arístegui, 2015). While suspended POM, is generally collected with Niskin bottles or stand-alone pumps, fast sinking POM is collected with sediment traps. POM is differentiated from DOM as the fraction of marine organic matter retained by a filter of a given pore size that usually range from 0.2 µm (polycarbonate, polysulfone, or aluminium filters) to 0.7 µm (typically glass fiber filters; Repeta, 2015). The CVFZ is characterized by the lateral export of organic matter from the coast to the open ocean not only in the surface nepheloid layer (SNL) but also in intermediate (INL) and bottom (BNL) nepheloid layers (Karakaş et al., 2006; Nowald et al., 2006; Fischer et al., 2009; Ohde et al., 2015), which affects carbon fluxes in the adjacent deep ocean waters (Alonso-González et al., 2009; Arístegui et al., 2020).

Information on the elemental composition of DOM, and suspended and sinking POM is crucial to understand the contrasting contribution of the three organic matter pools to the respiration in the ocean interior, which dictates the efficiency and magnitude of the biological carbon pump (Verdugo et al., 2004; Hopkinson and Vallino, 2005; Boyd et al., 2019). While an important effort has been made to characterize the origin, elemental composition and fluxes of sinking POM in the CVFZ (Helmke et al., 2005; Iversen et al., 2010; Álvarez-Salgado and Arístegui,

2015 and references therein; Fischer et al., 2019), the suspended POM and DOM fractions received much less attention (Alonso-González et al., 2009; Arístegui et al., 2020; Burgoa et al., 2020). Therefore the aim of this work is to study the dynamics of the dissolved and suspended fractions of organic matter in the CVFZ, and quantify the contribution of these pools to the remineralisation of organic matter in the dark ocean. Full-depth profiles of the carbon (C) and nitrogen (N) content of DOM and suspended POM were obtained in a box (17–23°N, 17–26°W) that embraced the giant filament of Cape Blanc and the CVF. Differences in dissolved organic carbon (DOC) and suspended particulate organic carbon (POC) stocks, C:N molar ratios, and their relative contribution to dissolved oxygen consumption and nitrate production following organic matter mineralisation between transects (Northern, NACW domain; Southern, SACW domain; Eastern, Coastal Transition Zone domain; and Western, open ocean domain) are examined for the surface, central, intermediate and deep water masses present in the study area.

2. Materials and methods

2.1. Sampling strategy

The FLUXES I cruise (R/V Sarmiento de Gamboa; Las Palmas - Las Palmas, 12 July to 9 August 2017) consisted of four transects (Northern, Western, Southern and Eastern) defining a box crossing the CVFZ, including 35 hydrographic stations 50 nautical miles apart (Fig. 1). Three types of stations were occupied: short (nocturnal), medium (diurnal) and long (full day), which lasted 3, 9 and 27 h, respectively (Fig. 1). Medium and long stations were sampled from the surface to 4000 db, whereas short stations were sampled down to 2000 db. A rosette sampler carrying 24 Niskin bottles of 12 L was used. It was equipped with a conductivity, temperature and depth (CTD) probe (SeaBird SBE911 plus), and sensors for dissolved oxygen (SeaBird SBE43), fluorescence of chlorophyll (SeaPoint SCF), and turbidity (SeaPoint STM). Sampling depths were selected based on the continuous records of the sensors attached to the rosette. Fifteen levels were sampled in medium and long stations (down to 4000 db) and 10 levels in short stations (down to 2000 db).

CTD conductivity was calibrated with water samples taken from the rosette and analysed on board with a Guildline 8410-A Portasal salinometer. Samples for dissolved oxygen (O_2) determination were analysed on board by the Winkler potentiometric method following the procedure described by Langdon (2010). The apparent oxygen utilization (AOU) was calculated with the equation of Benson & Krause, $AOU = O_{2sat} - O_2$ (UNESCO, 1986), where O_{2sat} is the dissolved oxygen

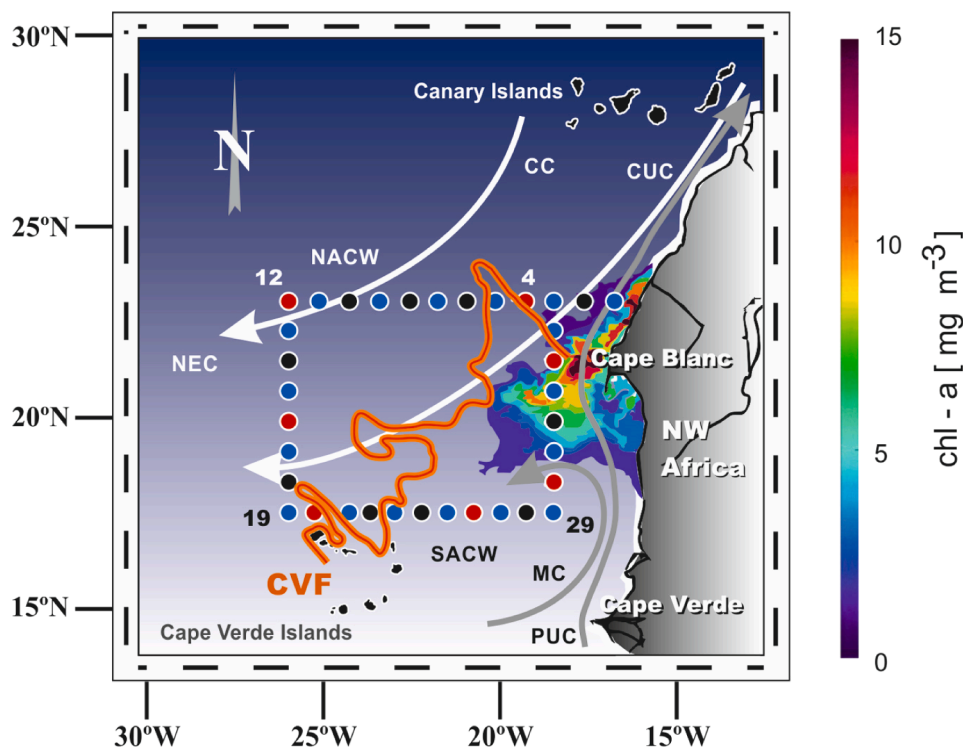


Fig. 1. Map of the FLUXES I cruise. The dots represent the stations (blue: diurnal stations; black: nocturnal stations; red: long stations) the numbers indicate the station number, from 1 to 35. The main geographic and oceanographic features and currents are represented. The white arrows correspond to the Canary Current (CC), Canary Upwelling Current (CUC) and North Equatorial Current (NEC), the grey arrows to the Mauritania Current (MC) and the Poleward Undercurrent (PUC) (taken from Pelegrí and Peña-Izquierdo, 2015). The orange line represents the approximate position of the CVF during the cruise (taken from Burgoa et al. in review), and the colour contours near the coast represent the satellite chlorophyll *a* associated with the Cape Blanc filament on 24/07/2017 (from Copernicus Marine Service). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

saturation at local TS. The chlorophyll (Chl) fluorescence sensor was calibrated with water samples taken at 4 depths in the photic layer at each medium and long stations, where 500 mL of water was filtered and phytoplankton cells concentrated on 25 mm GF/F filters for chlorophyll *a* determination. Pigments were extracted with 10 mL of 90% acetone at 4 °C in the dark for 24 h. The extracts were estimated fluorometrically by means of a Turner Designs bench fluorometer 10-AU, previously calibrated with pure chlorophyll *a* (Sigma Co.), as described in Holm-Hansen et al. (1965).

Water samples for the determination of inorganic nutrient concentrations were taken directly from the Niskin bottles in 25 mL polyethylene vials that were frozen at −20 °C until analysis in the base laboratory. A similar procedure was followed for dissolved organic carbon/total dissolved nitrogen (DOC / TDN) samples, which were collected in 30 mL amber glass vials with Teflon stoppers. Although these samples were not filtered, they will be named as DOC and TDN, because they mainly represent DOM due to the low concentrations of POM in oceanic waters. For the analysis of the concentration of suspended particulate organic carbon and nitrogen (POC and PON), 2–5 L of water were filtered onto pre-combusted (450 °C, 4 h) 25 mm Whatman GF/F filters in a filtration system with a mild vacuum (pressure difference < 300 mm Hg). Filters were transferred to 2 mL Eppendorf vials and dried for 12 h in a vacuum desiccator with silica gel. After drying, they were frozen at −20 °C until analysis in the base laboratory. Dissolved oxygen, inorganic nutrients, DOC/TDN and POC/PON were collected at all stations and all sampling depths.

2.2. Analytical determinations

Micromolar concentrations of nutrient salts (nitrate, nitrite, ammonium, phosphate and silicate) were determined simultaneously by segmented flow analysis in an Alliance Futura autoanalyser following the methods of Hansen and Koroleff (1999) except for the case of ammonium, which was measured by the fluorometric method of Kérouel and Aminot (1997). Frozen samples were gently melted in the dark overnight and vigorously shaken before analytical determination.

The determination of suspended POC and PON in 25 mm GF/F filters

was carried out by high temperature catalytic oxidation at 900 °C in a Perkin Elmer 2400 elemental analyser. At two selected stations, one from the Eastern (coastal) and one from the Western (open ocean) transects, the filters were divided into two parts; one part was measured directly and the other part was exposed to HCl fumes for 24 h to remove CaCO₃ before determination. After checking that CaCO₃ represented < 10% of carbon in these filters, the rest of the filters were analysed directly. Therefore, although we have measured total carbon, we will refer to these measurements as POC because of the low levels of CaCO₃.

DOC and TDN were analysed by high temperature catalytic oxidation at 680 °C with a Shimadzu TOC-V analyser connected in line with a TNM1 measuring unit. After melting, the samples were acidified to pH < 2 and degassed with high purity N₂ to eliminate the CO₂ before measurement. Aliquots of 150 µL were injected in the analyser. Measurements were made in triplicate to quintuplicate. The accuracy was checked daily with the DOC/TDN reference materials (CRM) provided by D.A. Hansell (University of Miami, USA). Measured concentrations of the CRM were $44.9 \pm 1.7 \mu\text{mol C L}^{-1}$ ($n = 10$) and $31.7 \pm 1.4 \mu\text{mol N L}^{-1}$ ($n = 10$); the certified values are 43–45 $\mu\text{mol C L}^{-1}$ and 32–33 $\mu\text{mol N L}^{-1}$ (DSW Batch 16–2016*Lot 05–16). DON concentrations were calculated as $\text{DON} = \text{TDN} - \text{DIN}$ where DIN is the dissolved inorganic nitrogen (nitrate + nitrite + ammonium).

2.3. Water mass analysis

An optimum multiparameter (OMP) inverse method (Karstensen and Tomczak, 1998) has been used for the quantification of the water types (WTs) that contribute to the water samples collected in the CVFZ during FLUXES I. A water type (WT) is a body of water formed in a particular region of the ocean with certain thermohaline and chemical properties (Tomczak, 1999).

In our OMP, the variables used to define the WTs were potential temperature (θ), salinity (*S*), silicate (SiO₄H₄) and the conservative chemical parameter NO ($\text{NO} = \text{O}_2 + \text{R}_\text{N} \cdot \text{NO}_3^-$ (Broecker, 1974) where R_N was set to a constant value of 9.3 mol O₂ mol NO₃^{−1} (Anderson, 1995)), and represents the Redfield ratio of dissolved oxygen consumption to nitrate production during the mineralisation of biogenic

organic matter in the ocean. We identified 11 wt in the CVFZ: Madeira Mode Water (MMW); Eastern North Atlantic Central Water (ENACW) of 15 °C and 12 °C; South Atlantic Central Water of 18 °C and 12 °C; Subpolar Mode Water (SPMW); Antarctic Intermediate Water (AA); Mediterranean Water (MW); Labrador Sea Water (LSW); and upper and lower North East Atlantic Deep Water (NEADW). The physical and chemical properties of most of these WTs were obtained from the literature (Table 1), except for AA and SACW of 18 °C and 12 °C, which were defined from the average characteristics of these WTs in the Eastern Equatorial Atlantic, between 5° N–5° S and 20–30° W (World Ocean Atlas, 2013). This source region was chosen because AA and SACW are transported from the Equatorial Atlantic to the CVFZ by the Mauritanian current (Fig. 1). The in situ values of θ , S , SiO_4H_4 and NO from the rosette sampler profiles collected during the cruise were used to calculate the contribution of each WT to every sample, by solving this set of mass balance equations per sample:

$$\sum_i X_{ij} \cdot \theta_i = \theta_j + R\theta_j$$

$$\sum_i X_{ij} \cdot S_i = S_j + RS_j$$

$$\sum_i X_{ij} \cdot (\text{SiO}_4\text{H}_4)_i = (\text{SiO}_4\text{H}_4)_j + R(\text{SiO}_4\text{H}_4)_j$$

$$\sum_i X_{ij} \cdot \text{NO}_i = \text{NO}_j + R\text{NO}_j$$

$$\sum_i X_{ij} = 1 + R \sum$$

where the subscript i corresponds to each WT and j to every sample; X_{ij} are the proportion of water type i in sample j ; θ_i , S_i , $(\text{SiO}_4\text{H}_4)_i$ and NO_i are the WT i values of θ , S , SiO_4H_4 and NO ; and θ_j , S_j , $(\text{SiO}_4\text{H}_4)_j$ and NO_j are the values for each variable in sample j ; R are the residuals each the mass balance equations and for the mass conservation equation for sample j .

These linear mixing equations were normalized and weighted. For

Table 1

Thermohaline and chemical characteristics (average value $\pm$ uncertainty) of the water types (WT) introduced in the OMP analysis of the water masses present in the CVFZ. Determination coefficient (R^2), standard error (SE) of the estimate and number of samples (n) of the multiple-regression of the measured and back-calculated variables.

WT	θ_i (°C)	S_i	SiO_4H_4 ($\mu\text{mol kg}^{-1}$)	NO_i ($\mu\text{mol kg}^{-1}$)
SACW ₁₈ ^a	18.03 $\pm$ 0.05	35.82 $\pm$ 0.03	5.1 $\pm$ 0.1	260 $\pm$ 8
MMW ^b	20.0 $\pm$ 0.5	37.00 $\pm$ 0.04	0.4 $\pm$ 0.3	225 $\pm$ 10
ENACW ₁₅ ^b	15.3 $\pm$ 0.4	36.10 $\pm$ 0.02	2.2 $\pm$ 1.7	264 $\pm$ 8
SACW ₁₂ ^a	12.0 $\pm$ 0.1	35.16 $\pm$ 0.01	10.8 $\pm$ 0.1	345 $\pm$ 7
ENACW ₁₂ ^c	12.2 $\pm$ 0.4	35.66 $\pm$ 0.02	4.9 $\pm$ 0.2	322 $\pm$ 8
SPMW ^c	8.2 $\pm$ 0.4	35.23 $\pm$ 0.01	14.5 $\pm$ 0.4	386 $\pm$ 7
AA ^a	4.89 $\pm$ 0.03	34.51 $\pm$ 0.02	30.3 $\pm$ 0.8	464 $\pm$ 8
MW ^c	11.8 $\pm$ 0.1	36.50 $\pm$ 0.01	7.2 $\pm$ 0.7	304 $\pm$ 9
LSW ^c	3.4 $\pm$ 0.2	34.89 $\pm$ 0.12	19.5 $\pm$ 0.4	446 $\pm$ 9
UNEADW ^d	2.500 $\pm$ 0.003	34.940 $\pm$ 0.003	34.8 $\pm$ 0.3	447 $\pm$ 7
LNEADW ^d	1.980 $\pm$ 0.002	34.884 $\pm$ 0.003	44.4 $\pm$ 0.3	462 $\pm$ 7
R^2	0.999	0.999	0.988	0.977
SE	0.04	0.005	1.2	11
N	307	307	307	307

^a This work, taken from the WOA13 in the Equatorial Atlantic.

^b Álvarez and Álvarez-Salgado (2009); Lønborg and Álvarez-Salgado (2014).

^c Pérez-Rodríguez et al. (2001); Álvarez and Álvarez-Salgado (2009).

^d Pérez-Rodríguez et al. (2001); Lønborg and Álvarez-Salgado (2014).

normalization, the mean and standard deviation values of the parameters in Table 1 were used. The equations were weighted considering the measurement error of each parameter in relation to its variability in the study area and its relative conservative nature. Weights of 10, 10, 2 and 1 were assigned to θ , S , NO and SiO_4H_4 , respectively. The weights assigned to the NO and SiO_4H_4 mass balance equations were lower because both variables are affected by non conservative processes: SiO_4H_4 is influenced by silica dissolution and the R_N of NO is variable, which introduces more uncertainty in the mass balance equations. We assumed that the mass is accurately conserved, so a weight of 100 was assigned to the mass conservation equation.

Since we have 5 mass balance equations and 11 WTs, in order to solve the OMP it is necessary to define oceanographically consistent WT mixing groups, taking into account that mixing occurs with adjacent WTs along or across isopycnals and a maximum of 4 WTs can mix simultaneously (Álvarez et al., 2014). The 5 groups of WTs defined, from surface to 4000 db were: 1) MMW - SACW18 - ENACW15; 2) SACW18 - ENACW15 - SACW12 - ENACW12; 3) SACW12 - ENACW12 - SPMW - AA; 4) SPMW - AA - MW - LSW; and 5) MW - LSW - LNEADW - UNEADW (Fig. 2). Each sample was assigned to a particular group according to its θ and S , and then the system of mass balance equations was solved for each sample.

The reliability of the OMP analysis was assessed with the determination coefficient (R^2) and standard error (SE) of the residuals of the linear regression between measured and back-calculated variables (θ , S , SiO_4H_4 and NO) (Álvarez et al., 2014). Note that our OMP reproduced the thermohaline and chemical fields during the cruise quite accurately, as demonstrated by the high R^2 (>0.977) and low SE of the linear regression of the measured and predicted values of θ , S , SiO_4H_4 and NO (Table 1).

Samples located between the surface and the base of the deep chlorophyll maximum (DCM) were excluded from the OMP analysis because the heat, freshwater and gas exchange with the atmosphere and the non conservative behaviour of O_2 and nutrient salts would disprove the assumption of conservation of the mass balance equations. Therefore, the OMP was applied to 307 of the 419 samples in which DOC, DON, PON and POC were determined.

2.4. WT Proportion-weighted average values

The WT proportion-weighted average value of any variable (N) in a given WT, henceforward archetype concentration of N , can be calculated using the X_{ij} obtained as detailed in Section 2.3, and the concentration of N in each sample (Álvarez-Salgado et al., 2013). The equation is:

$$N_i = \frac{\sum_j X_{ij} \cdot N_j}{\sum_j X_{ij}}$$

where N_i is the archetype value of N in water mass i ; X_{ij} is the proportion of WT i in sample j and N_j is the value of N in sample j . The standard error of the archetype value is obtained as:

$$SE_{N_i} = \frac{\sqrt{\sum_j X_{ij} (N_j - N_i)^2}}{\sum_j X_{ij}}$$

Finally, the proportion of the total volume of water sampled during the cruise that correspond to a given water mass (%VOL _{i}) can be calculated as:

$$\%VOL_i = 100 \cdot \frac{\sum_j X_{ij}}{n}$$

where n is number of samples. In this study, we have calculated archetype values of Z , S , θ , AOU, DOC, DON, POC and PON for the 11 WTs identified in the study area. Moreover, N_i values were calculated also for the Northern (stns 3 to 12, excluding stations 1 and 2 over the

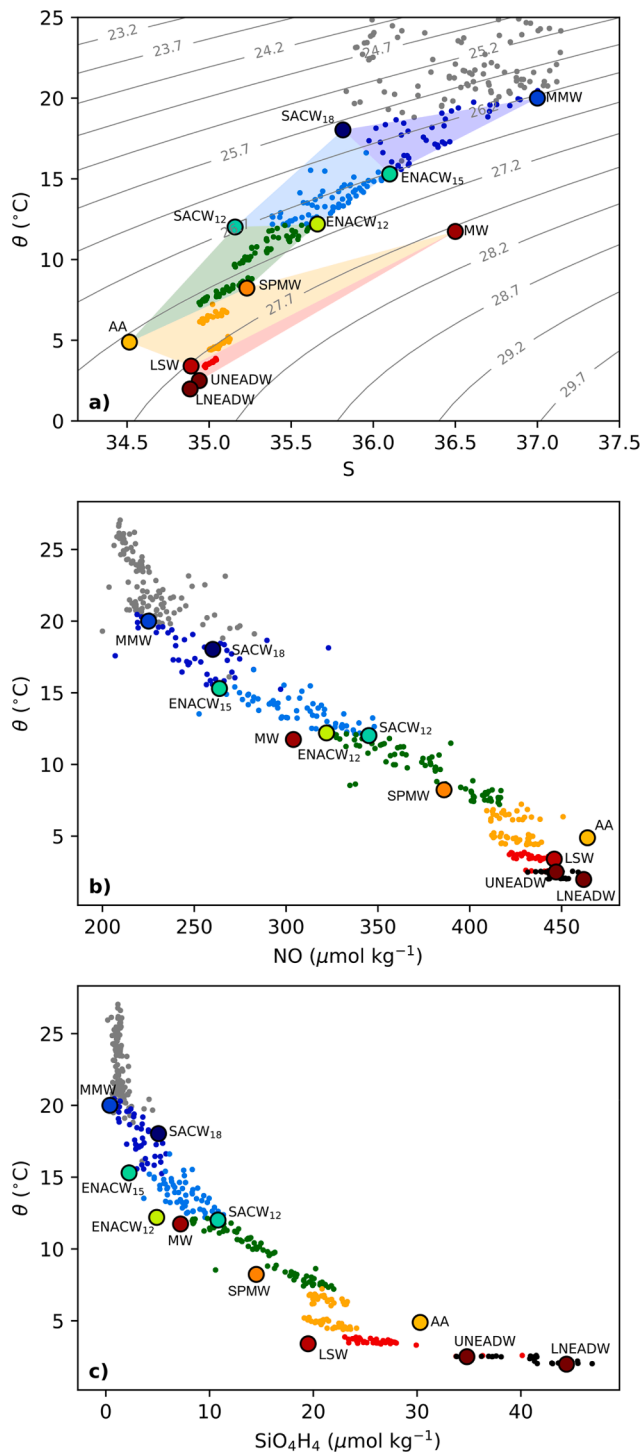


Fig. 2. Potential temperature (θ) versus salinity (a), NO (b) and SiO_4H_4 (c) for FLUXES I. Dots represent the water samples collected and the shading colours in (a) identify the mixing group assigned to each sample. Potential density contours are also shown in (a).

continental shelf) and Southern transects (stns 19 to 29) -separated by the CVF- and the Western transect (stns 12 to 19) -located in the open ocean- and the Eastern transect (stns 29 to 35 and stn 3) near the coast (Fig. 1), to assess geographical differences between transects within a given WT.

2.5. Multiple regression models

The fraction of the total variability of the distribution of a non-conservative variable (e.g. DOC, DON, POC, PON) explained by WT mixing can be calculated from a multiple linear regression of the non-conservative variable with the water mass proportions:

$$N_j = \sum_i X_{ij} \cdot n_i$$

where N_j is the concentration of the non-conservative variable, with $j = 1 \dots n$ (one per sample) and n_i is the correlation coefficient of WT i , with $i = 1 \dots m$ (one per WT). The per cent of explained variability is indicated by the determination coefficient (R^2) and the explanatory power by the standard error of the estimate (SE) of the multiple regression.

However, the distributions of DOC, DON, POC or PON depend not only on the conservative mixing of WTs but also on the biogeochemical processes that occur alongside mixing. The main biogeochemical process in the ocean interior is organic matter mineralisation and it can be traced with AOU (Álvarez et al., 2014). The observed organic matter mineralisation is the result of the large-scale mineralisation from the formation area of each WT to the study area and the local mineralisation that occurs within the study area (Álvarez-Salgado et al., 2013; Álvarez et al., 2014), the CVFZ in our case. On the one hand, the large-scale mineralisation is assumed to be included in the archetype value of each WT and, therefore, accounted for by the multiple linear regression with X_{ij} . On the other hand, a multiple linear regression with X_{ij} and AOU (for DOC, DON, POC and PON) or NO_3^- (for DON and PON) would quantify the local mineralisation of DOM and suspended POM:

$$N_j = \sum_i X_{ij} \cdot n_i + \beta \cdot \text{AOU (or } \text{NO}_3^-)$$

In this multiple linear regression of n linear equations with $m + 1$ unknowns, the extra unknown (β) is the coefficient that relates DOC, DON, POC or PON with AOU or NO_3^- and represents the rate of change of DOM and suspended POM with AOU or NO_3^- that does not depend on WT mixing (Álvarez-Salgado et al., 2013). Therefore, when calculating the multiple linear regression of DON or PON with X_{ij} and NO_3^- , the coefficient β would represent the contribution of DON or PON to the local NO_3^- production by organic matter mineralisation. On the other hand, the coefficient β in the multiple linear regression of DOC or POC to AOU would represent the contribution of DOC or POC to the oxygen utilisation. In this case, once β is multiplied by the Redfield ratio of dissolved oxygen consumption to organic carbon mineralisation of 1.4 mol O_2 mol C^{-1} (Anderson, 1995), it would represent the contribution of DOC or POC to the local production of inorganic nitrogen by organic matter mineralisation. Finally, once the contribution of DOM and suspended POM to organic matter mineralisation is calculated from β , the contribution of sinking POM can be estimated indirectly by difference.

2.6. Epipelagic layer

As indicated above, samples collected in the epipelagic layer were excluded from the OMP analysis because of the non-conservative behaviour of θ , S, SiO_4H_4 and NO. Those samples have been used to study the impact of the CVFZ on the biogeochemistry of the epipelagic layer during the FLUXES I cruise. To do that, we have calculated average concentrations of the different variables (Z, S, θ , AOU, Chl-a, NO_2^- , NO_3^- , NH_4^+ , DOC, DON, POC and PON) at the surface (5 db), and the deep chlorophyll maximum (DCM), as well as vertically-averaged from the surface to the base of the DCM. Vertical averages considered the uneven vertical sampling spacing by using the trapezoid rule to weight each sample. The average values were obtained for the whole cruise as well as for each individual transect. In the Northern transect, the first two stations were excluded because they are outside the box, and due to their position close to the continental shelf, they are affected by distinct

processes from those observed in the oceanic waters, biasing the calculation of the average values of the Northern transect (Fig. 1).

3. Results

3.1. Hydrographic setting of the CVFZ

3.1.1. Epipelagic waters

The relative position of the four transects with respect to the CVF during FLUXES I can be documented using the isohaline that defines the

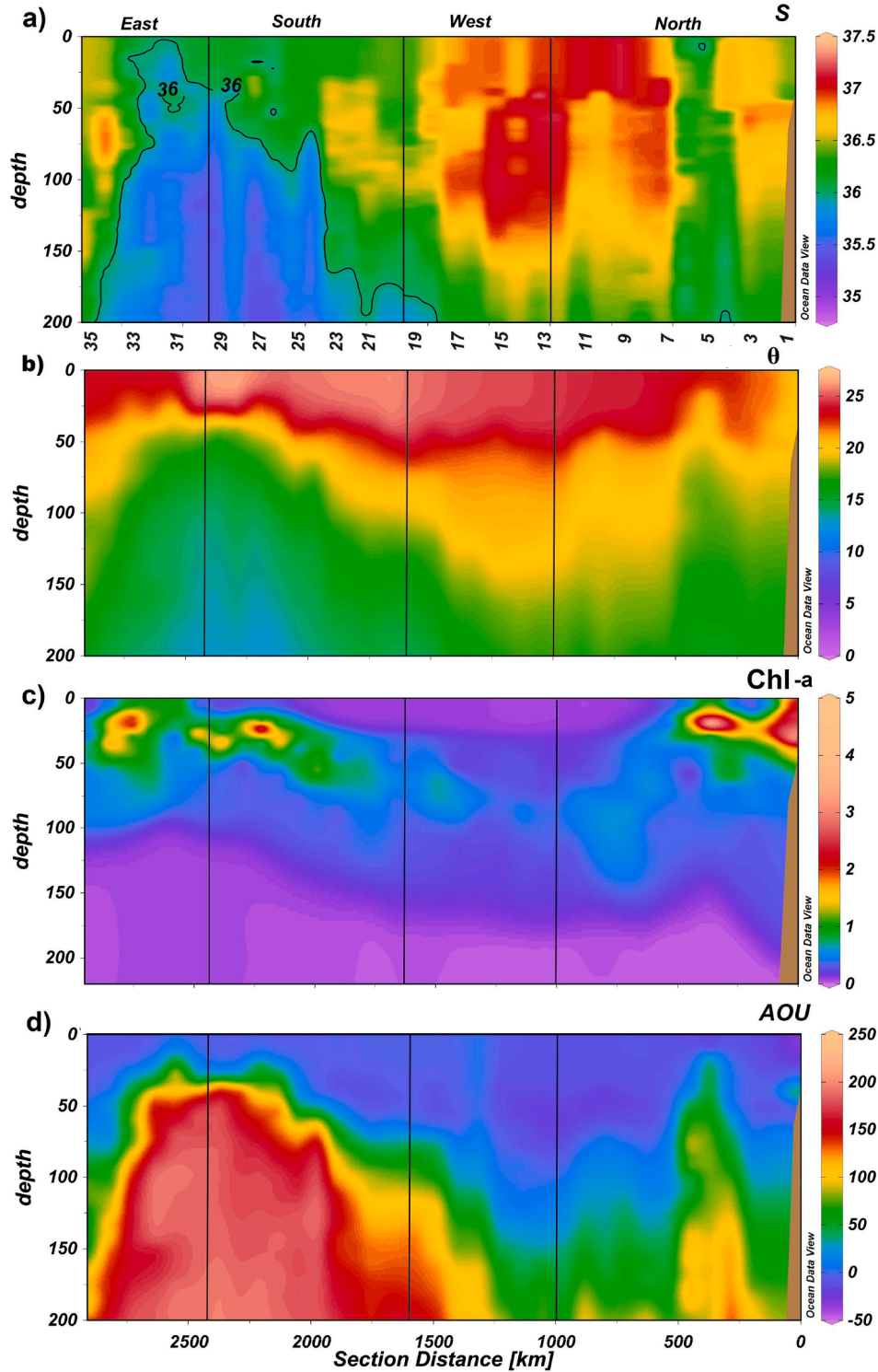


Fig. 3. Distribution of salinity (S) (a), potential temperature (θ) in $^{\circ}\text{C}$ (b), Chl (chlorophyll a) in $\mu\text{g L}^{-1}$ (c) and apparent oxygen utilization (AOU) in $\mu\text{mol kg}^{-1}$ (d), in the epipelagic layer during the FLUXES I cruise. The black line in (a) represents the 36 isohaline defining the location of the CVF, and the vertical black lines represent the corners of the FLUXES hydrographic box. The y-axis can be read as station number (a) or section distance (d). Produced with Ocean Data View (Schlitzer, 2017).

front (36 at 150 db) (Zenk et al., 1991) (Fig. 3a). According to this criterion, the Western and Northern transects were located to the North of the front, while the front intersected the Southern and Eastern transects around stations 34 and 24, respectively (Fig. 1). Due to the dominance of saline waters of North Atlantic origin to the North of the front, the average salinity was significantly higher in the Northern and Western transects (36.7–36.8), than in the Southern and Eastern transects (36.2) (Fig. 3a, Table S1). Mesoscale variability was super-imposed over this large-scale pattern, notably due to the presence of a narrow intrusion of relatively cold and less saline water of South Atlantic origin centred at stn 5 in the Northern transect. The mean depth-averaged temperature down to the base of the DCM was 22.3 °C with non-significant differences between transects (Fig. 3b, Table S1). Nonetheless, the Southern transect was characterized by significantly higher ($p < 0.0005$) surface (5 db) temperatures (26.1 °C) than the Northern transect (23.9 °C).

The DCM was shallower and with higher Chl-a levels towards the coast (Fig. 3c). The average depth of the DCM and average Chl-a levels in the Northern and Southern transect, were similar ($p > 0.3$) while the Western (oceanic) and Eastern (coastal) transects showed the highest

differences, being the DCM deeper and with lower concentrations in the Western transect (Table S1).

The distribution of AOU (Fig. 3d) matched the predominant form of inorganic nitrogen, NO_3^- (Fig. 4a), increasing significantly with depth (Table S1). In the same way as salinity and temperature, marked differences were observed between the Northern and Southern transects of the CVF: depth-averaged concentrations of AOU and NO_3^- were significantly lower ($p < 0.05$) in the Northern and Western transects (< 4.5 and $< 2.3 \mu\text{mol kg}^{-1}$, AOU and NO_3^- respectively) than in the Southern and Eastern transects (> 20 and $> 4.6 \mu\text{mol kg}^{-1}$, respectively). The intrusion of water with South Atlantic origin at stn 5 is traced by higher AOU and NO_3^- values in the upper 100 db (Figs. 3d, 4a).

NH_4^+ (Fig. 4b) and NO_2^- (Fig. 4c) were, in general, very low (< 0.06 and $< 0.14 \mu\text{mol kg}^{-1}$), except in the surroundings of the DCM. An average NH_4^+ concentration of $0.09 \pm 0.03 \mu\text{mol kg}^{-1}$ was observed at the DCM, with significantly ($p < 0.01$) lower values ($0.02 \pm 0.03 \mu\text{mol kg}^{-1}$) in the Western (oceanic) transect and higher values ($0.19 \pm 0.10 \mu\text{mol kg}^{-1}$) in the Eastern (coastal) transect. The average concentration of NO_2^- at the DCM was $0.23 \pm 0.03 \mu\text{mol kg}^{-1}$, being also significantly

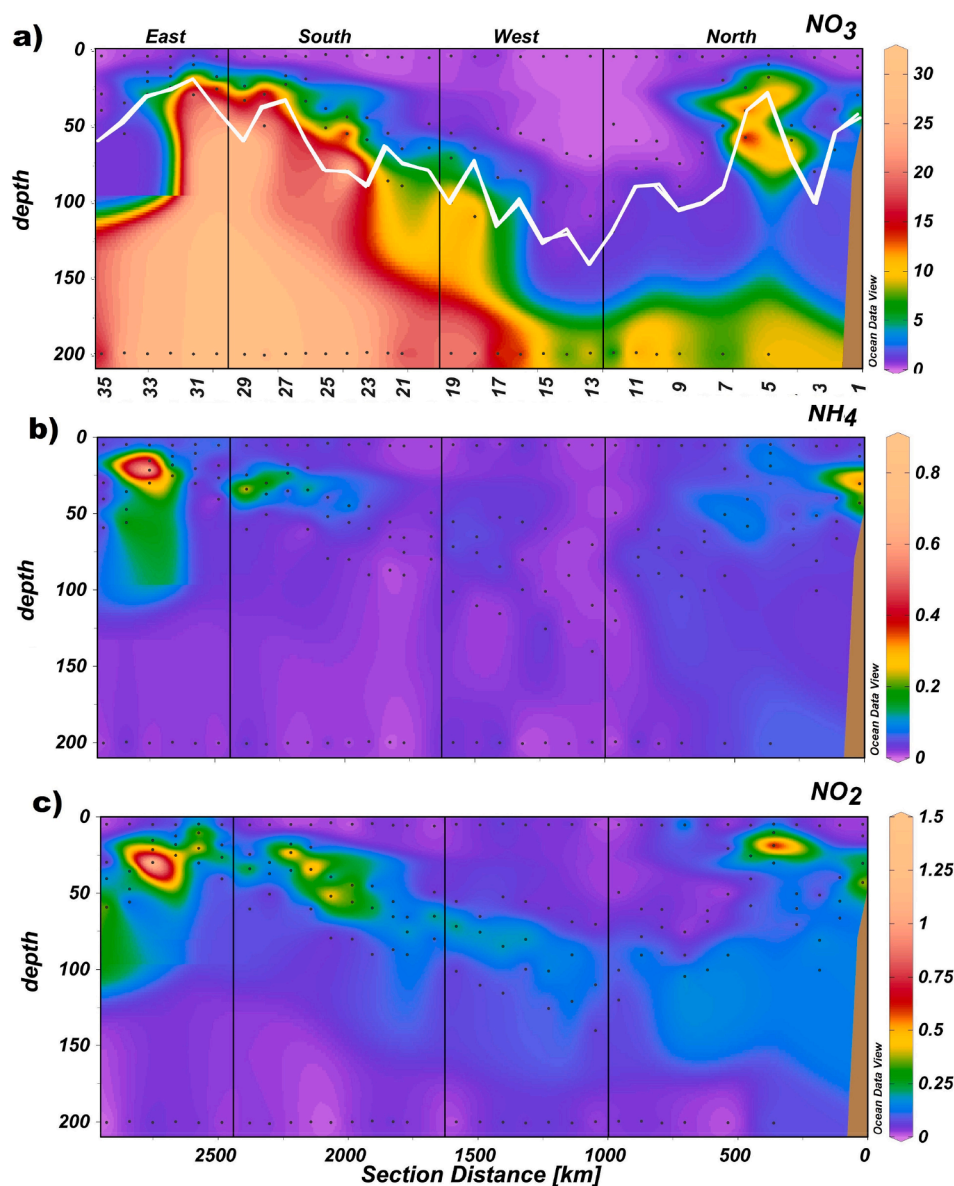


Fig. 4. Distributions of nitrate (NO_3^-) in $\mu\text{mol kg}^{-1}$ (d) (a), ammonium (NH_4^+) in $\mu\text{mol kg}^{-1}$ (b), and nitrite (NO_2^-) in $\mu\text{mol kg}^{-1}$ (c), in the epipelagic layer during the FLUXES I cruise. Dots represent samples, vertical back lines represent the corners of the FLUXES I hydrographic box and the white line in panel a represents the base of the DCM. The y-axis can be read as station number (a) or section distance (c). Produced with Ocean Data View (Schlitzer, 2017).

lower in the Western transect ($p < 0.025$).

NH_4^+ and NO_2^- also showed relatively elevated concentrations at stn 5 in the Northern transect, coinciding with the relative maxima in temperature, Chl-a, AOU and NO_3^- , and minimum in salinity, reflecting the influence of the mesoscale variability of the CVF in the distribution of biogeochemical properties.

3.1.2. Meso- and bathypelagic waters

Below the epipelagic layer, in the depth range between 100 and 700 db, hydrographic variability was governed by the distribution of the central water types. SACW_18 and MMW, characterized by potential temperatures of 18 °C and 20 °C, respectively (Table 1, Fig. 5ab, S1a), and centred at about 110 db (Table 2) were the shallowest central water types. SACW_18 and MMW were defined by distinct salinity values, 35.82 and 37.00, respectively (Table 1, Fig. 5ab, S1b), as expected for two water masses of contrasting North and South Atlantic origin, and represented 4.78% and 5.48% of the total sampled water volume (Table 2) or 9.9% and 11.4% of the volume of central waters. ENACW_15 was located deeper, centred at 240 ± 14 db (Table 2) and represented 12.54% of the total sampled volume or 26.1% of the volume of central waters. Most of the ENACW_15 was present in the Northern and Western transects (42.8% and 25.0% of the volume of this WT, respectively), i.e. to the North of the CVF (Fig. 5c). SACW_12 and ENACW_12 were centred at 336 $\pm$ 28 db and 442 $\pm$ 19 db and were defined by a similar temperature of ~ 12 °C, but different values of salinity (35.16 and 35.66), SiO_4H_4 (10.8 and 4.9 $\mu\text{mol kg}^{-1}$) and NO (345 and 322 $\mu\text{mol kg}^{-1}$) (Table 1). ENACW_12 was the most abundant WT sampled during the cruise with 18.09% (Table 2, Fig. 5b) or 37.6% of the volume of central waters and was present in the four transects (from a minimum of 17.8% of the total volume in the Western transect to a maximum of 28.6% in the Northern transect). Conversely, SACW_12 occupied a volume of 7.25% (or 15.1% of the central waters) and was mostly present in the stations of the Southern transect, where 57% of the volume of this WT was found (Table 2, Fig. 5a). Again, in the Northern section, around stn 5, SACW_18 appears, although ENACW dominated in the rest of the transect (Fig. 5).

The intermediate water domain, approximately between 700 db and 1500 db, was occupied by AA, SPMW and MW. These water masses, of contrasting sub-Antarctic, sub-Arctic and Mediterranean origin, have remarkable differences in θ , S, SiO_4H_4 and NO (Table 1). It is particularly noticeable the salinity minimum observed in the Southern transect at about 1000 db, which corresponds to AA (Fig. 5c, S1b). The shallowest intermediate WT was the SPMW, centred at 809 ± 32 db, with a temperature of 8 °C and representing 7.18% of the sampled volume (or 34.3% of the intermediate waters).

SPWM was followed by AA at a depth of 873 ± 54 db, which had the lowest salinity 34.52 but the highest content of silicates 30.3 $\mu\text{mol kg}^{-1}$ and occupied a volume of 10.62% (or 50.8% of the intermediate waters). This water mass was mostly present to the South of the CVF (59.0% of the volume sampled in the Northern and Western transects) while SPMW predominated to the North of the front (56.8% of the sampled volume in the Eastern and Southern transects) (Fig. 5d). The deepest intermediate WT, MW, centred at 1455 ± 118 db, was much less represented with 3.09% (14.8% of the intermediate waters) and was almost absent in the Eastern transect (only 3.1% of the volume).

Finally, the deep waters domain was composed by three water types with similar salinity, yet different silicate content (Fig. S1c) due to their contrasting origin. While LSW is of pure North Atlantic origin, LNEADW and UNEADW are mixed with Antarctic waters that confer their higher silicate concentration. LSW was defined by a silicate concentration of 19.5 $\mu\text{mol kg}^{-1}$, it was located at 1648 ± 46 db and represented 15.58% of the sampled volume (50.3% of the deep waters), being the second predominant water mass sampled during FLUXES I. LSW was found in the four transects, ranging from 19.9% of the total volume in the Eastern transect to 29.4% in the Northern transect (Fig. 5a). In contrast, UNEADW was located deeper at 2742 ± 93 db and had a silicate

concentration of 34.8 $\mu\text{mol kg}^{-1}$, with a volume sampled of 11.72% (37.9% of the deep waters). LNEADW was placed at 3824 ± 101 db, with a silicate concentration of 44.4 $\mu\text{mol kg}^{-1}$ and a small representation of sampled volume of 3.66% (or 11.8% of deep waters). UNEADW was distributed in the four transects, ranging from 18.7% of the total volume in the Eastern transect to 29.0% in the Southern transect (Fig. 5d). In contrast, LNEADW concentrated in the Northern and Western transects (75% of the total volume of this WT) while it was practically absent in the Eastern transect. Bottom depth is the likely reason behind this difference (Fig. 5c).

3.2. Biogeochemical variability in epipelagic, meso- and bathypelagic waters of the CVFZ

3.2.1. Organic matter variability in epipelagic waters

DOC (Fig. 6a) and DON (Fig. 6b) showed maximum average concentrations of 80.9 ± 1.7 $\mu\text{mol L}^{-1}$ and 6.7 ± 0.2 $\mu\text{mol L}^{-1}$ at the surface layer (Table S1), and decreased with depth. At the surface, DOC and DON levels were significantly higher in the Southern transect, while at the DCM, the highest DOC and DON concentrations were found in the Eastern (coastal) transect (Table S1, Fig. 6a,b). The average C:N molar ratio of DOM throughout the epipelagic layer was 12.0 ± 0.3 and showed similar values in all transects at the surface and the DCM ($p > 0.05$).

POC (Fig. 6c) and PON (Fig. 6d) concentrations showed similar distributions with a marked decrease below the DCM. Maximum concentrations of POC and PON were observed at the stations near the coast, in the Northern and Eastern transects.

Maximum POC coincided with the DCM in the Eastern (10.4 ± 2.0 $\mu\text{mol L}^{-1}$) transect, in the Western (2.8 ± 0.3 $\mu\text{mol L}^{-1}$) transect it was located at the surface, while in the Northern (6.9 ± 2.1 $\mu\text{mol L}^{-1}$) and Southern (5.7 ± 1.1 $\mu\text{mol L}^{-1}$) transects it was somewhat above the DCM. Maximum PON concentrations followed the same pattern (Fig. 6d). In the upper 50 db, stn 5 shows very high values of DOC, DON, POC and PON.

The average value of the C:N molar ratio of POM over the epipelagic layer, 8.1 ± 0.4 , was significantly lower than the average C:N molar ratio of DOM, 12.0 ± 0.3 (Table S1). Average C:N ratios of POM were significantly higher in the Northern and Western transects (>8.9) than in the Southern and Eastern transects (<7.3) ($p < 0.005$). In the Western, Southern and Eastern transects, minimum values of the C:N ratio of POM were associated with the DCM as for the case of the C:N ratio of DOM (Table S1).

3.2.2. Organic matter variability in meso- and bathypelagic waters

AOU can be used to trace organic matter mineralisation in the meso- and bathypelagic layers of the CVFZ (Fig. 7a). In the central waters, the lowest archetype AOU values (50 to 120 $\mu\text{mol kg}^{-1}$) were found in the shallowest WTs (MMW, ENACW_15 and SACW_18), while the highest archetype AOU (192 $\mu\text{mol kg}^{-1}$) corresponded to SACW_12 (Table 2). In intermediate and deep waters, AOU decreased significantly ($p < 0.0005$) with depth from an archetype value of 176 $\mu\text{mol kg}^{-1}$ in SPMW to 85 $\mu\text{mol kg}^{-1}$ in the LNEADW. WTs mixing explained 94% of the variability of AOU, but the SE of 12.9 $\mu\text{mol kg}^{-1}$ was larger than the measurement error of ~ 1 $\mu\text{mol kg}^{-1}$ (Table 2).

Regarding the geographical variability within WTs, ENACW_12, the only central WT with significant presence in the four transects, showed significantly ($p < 0.01$) lower AOU levels in the Northern transect (144 $\mu\text{mol kg}^{-1}$) and higher in the Southern transect (204 $\mu\text{mol kg}^{-1}$; Table S2). In intermediate waters, AOU in SPMW and AA was also significantly higher in the Southern transect ($p < 0.05$). In deep waters, archetype AOU levels in LSW did not differ between transects, while in UNEADW, they were significantly ($p < 0.025$) higher in the Southern and Eastern transects (90–96 $\mu\text{mol kg}^{-1}$) as compared with the Northern and Western transects (83 $\mu\text{mol kg}^{-1}$).

Archetype DOC values (Fig. 7b) ranged from 58.9 ± 2.5 $\mu\text{mol L}^{-1}$ in

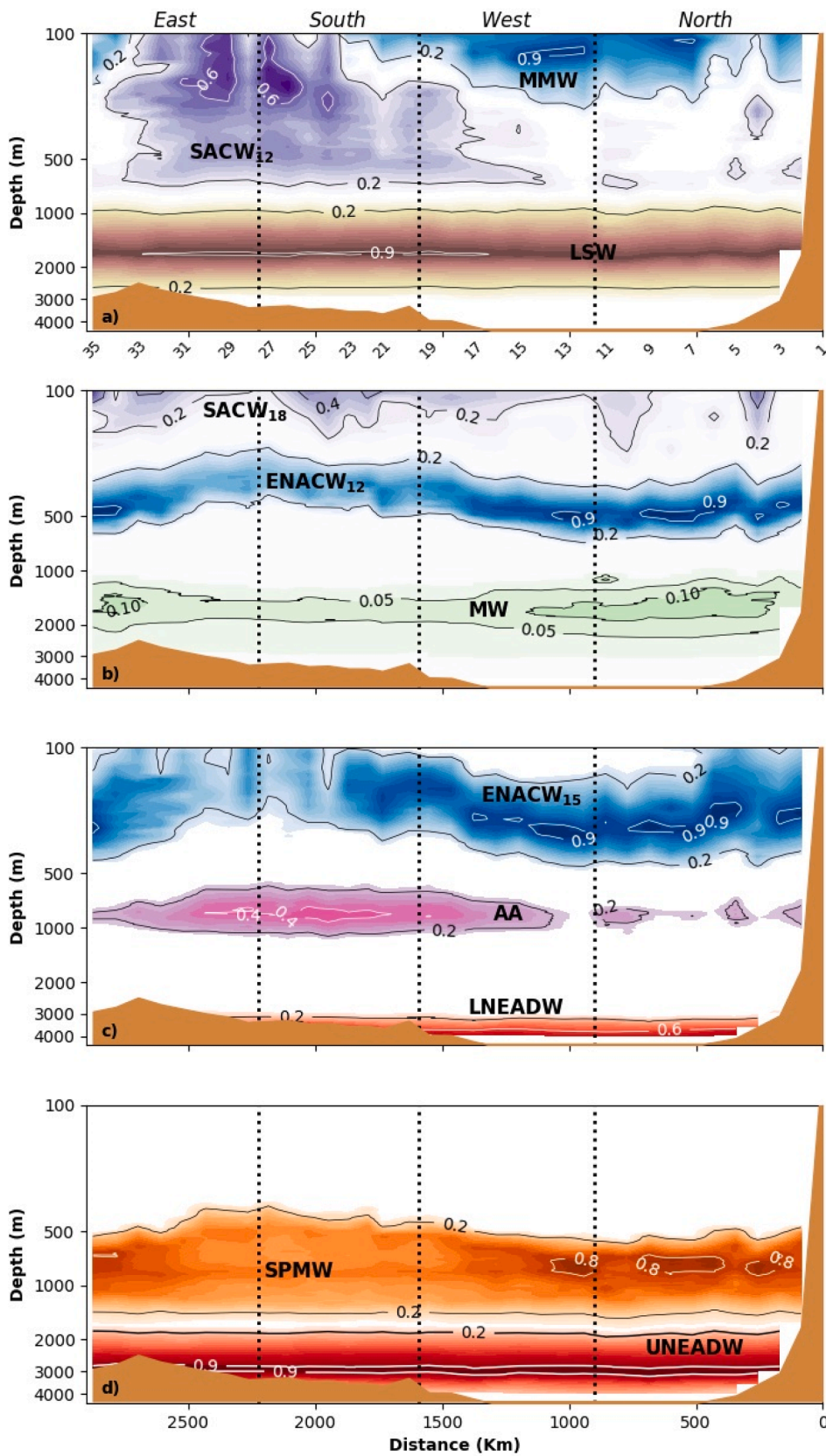


Fig. 5. Distributions of the water masses described in the CVFZ during the FLUXES I cruise. Vertical black lines represent the corners of the FLUXES I hydrographic box. Water mass proportions in this figure were derived with an OMP analysis applied to CTD data at 1 db vertical resolution. θ and S were measured directly with the CTD, while SiO_4H_4 and NO values were reconstructed by fitting the measured water sample concentrations to a non-linear combination of variables directly measured with the CTD (θ , S , O_2) (Figs. S1 and S2). The distribution of the water-masses among the panels was designed to avoid contour overlapping and does not follow an oceanographic criteria. The y-axis can be read as station number (a) or section distance in km (d).

Table 2

Percentage of the total volume of water sampled during FLUXES I that corresponded to each water type (%VOLI) and archetype values $\pm$ SE of depth (Zi), apparent oxygen utilization (AOUi), nitrate (NO_3^- i), dissolved organic carbon (DOCi), dissolved organic nitrogen (DONi), particulate organic carbon (POCi), particulate organic nitrogen (PONI), and C:N ratios of DOM and suspended POM for the water types intercepted during the cruise. Determination coefficient (R^2), standard error of the estimate (SE) of the multiple-regression of each variable with the water type proportions X_{ij} .

WT	%VOLI	Zi (db)	AOUi ($\mu\text{mol kg}^{-1}$)	NO_3^- i ($\mu\text{mol kg}^{-1}$)	DOCi ($\mu\text{mol L}^{-1}$)	DONi ($\mu\text{mol L}^{-1}$)
SACW_18	4.78%	102 $\pm$ 19	120.2 $\pm$ 15.4	16.9 $\pm$ 2.1	58.9 $\pm$ 2.5	4.6 $\pm$ 0.4
MMW	5.48%	113 $\pm$ 3	50.9 $\pm$ 10.5	6.5 $\pm$ 1.4	58.7 $\pm$ 1.6	5.1 $\pm$ 0.3
ENACW_15	12.54%	240 $\pm$ 14	117.8 $\pm$ 6.7	16.2 $\pm$ 1.0	52.3 $\pm$ 1.7	4.2 $\pm$ 0.3
SACW_12	7.25%	336 $\pm$ 28	191.7 $\pm$ 5.4	28.7 $\pm$ 1.0	49.9 $\pm$ 1.5	3.5 $\pm$ 0.3
ENACW_12	18.09%	442 $\pm$ 19	171.2 $\pm$ 4.8	27.2 $\pm$ 0.8	48.5 $\pm$ 0.6	3.4 $\pm$ 0.2
SPMW	7.18%	809 $\pm$ 32	175.7 $\pm$ 4.2	30.9 $\pm$ 0.7	44.8 $\pm$ 0.6	3.3 $\pm$ 0.3
AA	10.62%	873 $\pm$ 54	172.6 $\pm$ 5.7	31.3 $\pm$ 0.6	45.1 $\pm$ 0.6	3.1 $\pm$ 0.3
MW	3.09%	1455 $\pm$ 118	119.4 $\pm$ 9.2	25.9 $\pm$ 1.2	43.3 $\pm$ 0.9	3.0 $\pm$ 0.4
LSW	15.58%	1688 $\pm$ 46	104.0 $\pm$ 2.6	23.9 $\pm$ 0.3	43.4 $\pm$ 0.5	3.0 $\pm$ 0.2
UNEADW	11.72%	2742 $\pm$ 93	88.0 $\pm$ 1.1	22.2 $\pm$ 0.2	43.4 $\pm$ 0.6	2.9 $\pm$ 0.2
LNEADW	3.66%	3824 $\pm$ 101	85.1 $\pm$ 1.7	22.4 $\pm$ 0.2	43.5 $\pm$ 1.0	3.1 $\pm$ 0.3
R^2			0.94	0.95	0.63	0.44
SE			12.9	1.8	4.0	0.9

WT	POCi ($\mu\text{mol L}^{-1}$)	PONI ($\mu\text{mol L}^{-1}$)	C:N DOM (mol C mol N $^{-1}$)	C:N POM (mol C mol N $^{-1}$)
SACW_18	3.2 $\pm$ 1.2	0.33 $\pm$ 0.13	12.7 $\pm$ 1.1	9.9 $\pm$ 1.6
MMW	2.7 $\pm$ 1.0	0.25 $\pm$ 0.10	11.6 $\pm$ 0.7	10.9 $\pm$ 1.3
ENACW_15	1.5 $\pm$ 0.3	0.12 $\pm$ 0.03	12.4 $\pm$ 1.0	12.3 $\pm$ 0.9
SACW_12	1.4 $\pm$ 0.2	0.11 $\pm$ 0.02	14.4 $\pm$ 1.5	12.4 $\pm$ 1.4
ENACW_12	1.2 $\pm$ 0.1	0.09 $\pm$ 0.01	14.4 $\pm$ 1.1	12.5 $\pm$ 0.8
SPMW	0.79 $\pm$ 0.06	0.05 $\pm$ 0.01	13.6 $\pm$ 1.3	14.5 $\pm$ 1.3
AA	0.81 $\pm$ 0.06	0.06 $\pm$ 0.01	14.6 $\pm$ 1.3	14.2 $\pm$ 1.3
MW	0.70 $\pm$ 0.11	0.05 $\pm$ 0.01	14.4 $\pm$ 1.9	14.2 $\pm$ 2.2
LSW	0.67 $\pm$ 0.05	0.05 $\pm$ 0.01	14.5 $\pm$ 0.9	14.6 $\pm$ 1.1
UNEADW	0.68 $\pm$ 0.07	0.04 $\pm$ 0.01	15.2 $\pm$ 0.9	15.6 $\pm$ 1.3
LNEADW	0.71 $\pm$ 0.12	0.04 $\pm$ 0.01	14.8 $\pm$ 1.5	17.3 $\pm$ 2.0
R^2	0.26	0.29		
SE	1.3	0.13		

SACW_18 to $43.4 \pm 0.6 \mu\text{mol L}^{-1}$ in LNEADW (Table 2). The multiple regression with X_{ij} explained 63% of the DOC variability with a SE ($4.0 \mu\text{mol L}^{-1}$) that was about 3–4 times the measurement error of DOC. The addition of AOU to the multiple linear regression increased the explained variability to 70% and produced a 15% reduction of SE (Table 3). The AOU coefficient of the multiple regression, $-0.026 \pm 0.015 \text{ mol C mol O}_2^{-1}$, represents the average stoichiometric ratio of DOC to O_2 consumption for the meso- and bathypelagic waters in the CVFZ. This ratio is independent of water mass mixing and indicates that DOC mineralisation may explain $3.6 \pm 2.1\%$ of the inorganic carbon production in the area, assuming a $-\text{O}_2:\text{C}$ Redfield ratio of $1.4 \text{ mol O}_2 \text{ mol C}^{-1}$. The AOU coefficient of the multiple linear regression of DOC with X_{ij} and AOU was -0.061 ± 0.032 and $-0.150 \pm 0.038 \text{ mol C mol O}_2^{-1}$ for the Northern and Southern transects, respectively. Therefore, although DOC represented only $3.6 \pm 2.1\%$ of the organic matter

mineralised in the water masses of the CVFZ, it accounted for $9 \pm 4\%$ and as much as $21 \pm 5\%$ in the Northern and Southern transects, respectively (Table S3).

DON (Fig. 7c) decreased with depth from $5.1 \pm 0.3 \mu\text{mol L}^{-1}$ in MMW to $3.1 \pm 0.3 \mu\text{mol L}^{-1}$ in the deepest WT, LNEADW (Table 2). WTs mixing explained 44% of the variability of the distribution of DON in the CVFZ with a SE of $0.90 \mu\text{mol L}^{-1}$, 3 times the measurement error (Table 2). R^2 increased to 46% and SE reduced by 22% when the multiple linear regression included X_{ij} and AOU as explanatory variables. When NO_3^- was included as an explanatory variable, R^2 increased to 55% and SE reduced by 19%. In this case, the NO_3^- coefficient of -0.16 ± 0.02 indicated that 16% of nitrate is produced following DON mineralisation in the CVFZ, a larger fraction than obtained for DOC and AOU (Table 3). When analysing the DON to NO_3^- coefficient of the multiple linear regression with X_{ij} and NO_3^- transect by transect, we obtained significant values only for the Northern (-0.12 ± 0.03) and Southern (-0.16 ± 0.03) transects (Table S3).

POC (Fig. 7d) decreased with depth down to the MW level and then remained almost constant to the bottom. The highest concentrations of POC, 3.2 ± 1.2 and $2.7 \pm 1.0 \mu\text{mol L}^{-1}$, were found in the shallowest central waters SACW_18 and MMW, respectively (Table 2). WTs mixing explained only 26% of the POC variability in the CVFZ (Table 2), which increased to 38% when AOU was included in the multiple linear regression, accompanied by a SE reduction of 46% (Table 3). The AOU regression coefficient was $-0.032 \pm 0.005 \text{ mol C mol O}_2^{-1}$ (Table 3), indicating that suspended POC mineralisation may explain $4.48 \pm 0.70\%$ of the inorganic carbon produced following organic matter mineralization in the meso- and bathypelagic waters.

In central waters, no significant differences in POC concentration were found between transects, except for ENACW with significantly lower POC in the Western (oceanic) transect ($p < 0.025$) (Table S2, Fig. 7d). In intermediate waters, POC had a subtle increase in the Eastern (coastal) transect, whereas in deep waters a slight increase was associated to LSW. Per individual transects, the variability of POC explained with the multiple regression with X_{ij} and AOU was higher than the WT mixing model though with a small SE reduction of around 5% in the Northern and Southern transects, where POC represented only $6 \pm 2\%$ and $3 \pm 1\%$ of the inorganic carbon produced by organic matter mineralization during water mass mixing respectively (Table S3).

PON (Fig. 7e) generally mirrored the POC distribution in central and intermediate waters. The multiple regression of PON with X_{ij} explained 29% of its variance (Table 2) and the inclusion of AOU as explanatory variable increased the explained variance to 44% with a SE reduction of 23%, while inclusion of NO_3^- increased the explained variance to 34%. The AOU coefficient was -0.0034 ± 0.001 , while NO_3^- coefficient was -0.021 ± 0.004 , meaning that PON covers only $2.1 \pm 0.4\%$ of NO_3^- production following organic matter mineralisation (Table 3).

The lowest PON concentrations were found in the Western (oceanic) transect, while the Northern transect presented two PON maxima in SACW_18 and MMW (Table S2). In the intermediate and deep WTs, the highest concentrations were observed in the Eastern (coastal) transect ($p < 0.01$). The variability of PON explained with the multiple regression including X_{ij} and AOU or NO_3^- as explanatory variables was higher in the Southern than in the Northern transect (Table S3), with NO_3^- coefficients of -0.014 ± 0.002 and -0.039 ± 0.010 respectively, indicating that mineralised PON represented $1.4 \pm 0.2\%$ and $3.9 \pm 1.0\%$ of the NO_3^- produced.

Finally, the C:N molar ratio of POM increased significantly with depth from 9.9 ± 1.6 in SACW_18 to 17.3 ± 2.0 in LNEADW (Table 2). In central waters, the maximum C:N molar ratio of POM was observed in SACW_12 and ENACW_12, 12.4–12.5, while in the intermediate waters the ratio is similar for the three water masses, ranging from 14.2 to 14.5. In the deep waters, LNEADW presented the highest C:N molar ratio of POM. Concerning differences between transects, there were not significant differences in the C:N ratio of POM for ENACW_12 (Table S2). In the deep waters, the LSW presented the largest differences in C:N ratio

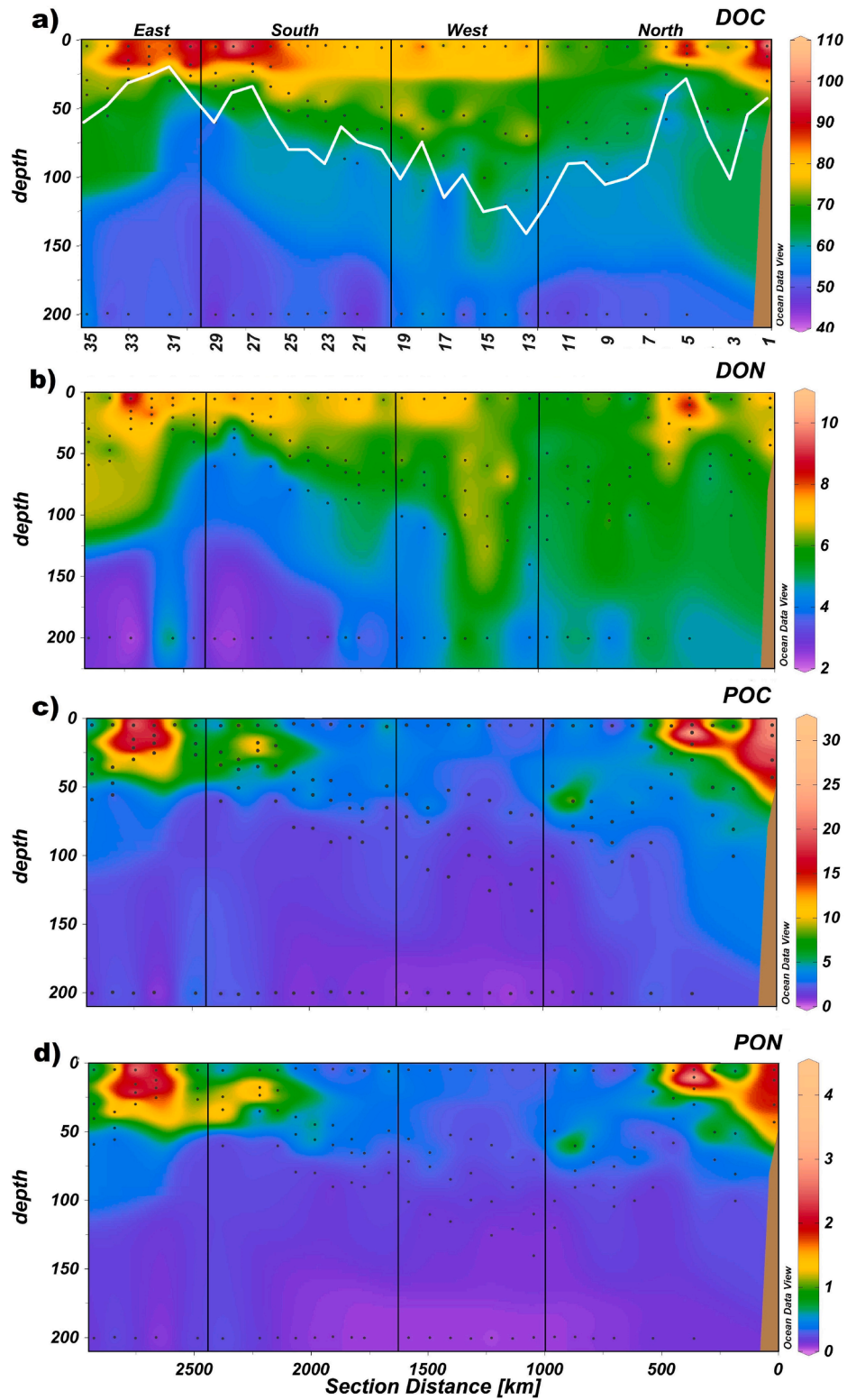


Fig. 6. Distributions of DOC in $\mu\text{mol L}^{-1}$ (a), DON in $\mu\text{mol L}^{-1}$ (b), POC in $\mu\text{mol L}^{-1}$ (c) and PON in $\mu\text{mol L}^{-1}$ (d) in the epipelagic layer during the FLUXES I cruise, dots represent samples, vertical black lines represent the corners of the FLUXES I hydrographic box, and the white line in a) represent the base of DCM. The y-axis can be read as station number (a) or section distance in km (d). Produced with Ocean Data View (Schlitzer, 2017).

($p < 0.0005$) being higher in the Western transect (Table S2).

4. Discussion

4.1. Variability in the epipelagic layer of the CVFZ

The relative position of the four transects in relation to the CVF and

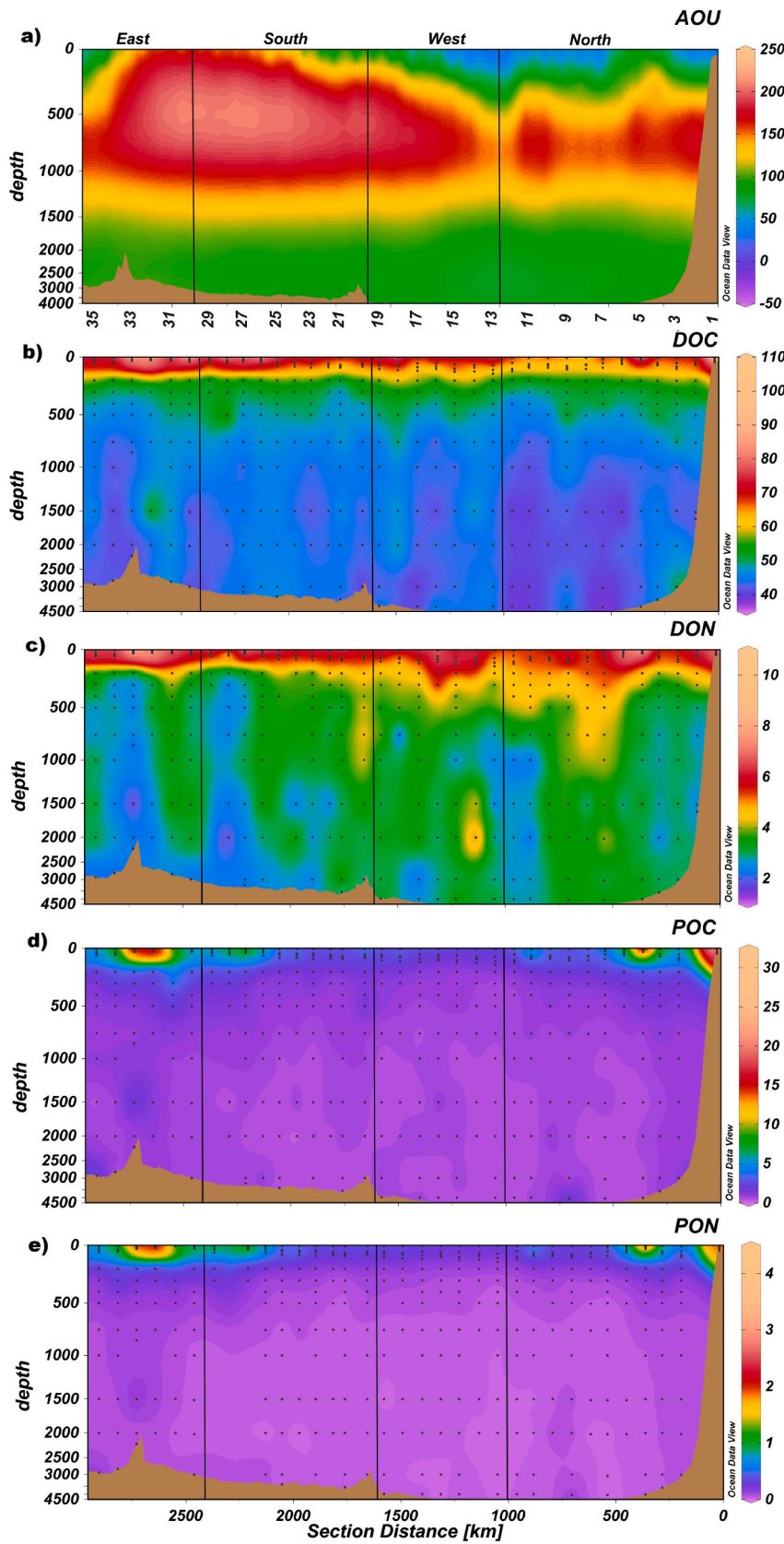


Fig. 7. Full depth distributions of AOU, in $\mu\text{mol kg}^{-1}$ (a), DOC in $\mu\text{mol L}^{-1}$ (b), DON in $\mu\text{mol L}^{-1}$ (c), POC in $\mu\text{mol L}^{-1}$ (d) and PON in $\mu\text{mol L}^{-1}$ (e) during the FLUXES I cruise. Dots represent samples (AOU measurements were obtained at 1 db vertical resolution) and vertical black lines represent the corners of the FLUXES I hydrographic box. Note that depth scale is not linear. The y-axis can be read as station number (a) or section distance in km (e). Produced with Ocean Data View (Schlitzer, 2017).

Table 3

Multiple linear regressions of DOC, DON, POC and PON with the water type proportions (X_{ij}) and AOU or NO_3^- . Determination coefficient (R^2_{+1}), standard deviation (SE_{+1}), percentage of R^2 increase ($\%R^2$) and percentage of SE reduction ($\%\text{SE}$) between the multiple linear regression with X_{ij} (Table 2) and the multiple linear regression with X_{ij} and AOU or NO_3^- , regression coefficient of each variable with AOU or NO_3^- (β), and standard error of the estimate (SE_β) and p -value of β .

	DOC vs. AOU	DON vs. AOU	POC vs. AOU	PON vs. AOU	DON vs. NO_3^-	PON vs. NO_3^-
R^2_{+1}	0.70	0.46	0.38	0.44	0.50	0.34
SE_{+1}	3.4	0.7	1.0	0.10	0.73	0.13
$\%R^2$	11	5	46	52	14	17
$\%\text{SE}$	15	22	23	23	19	0
β	-0.0260	-0.0130	-0.0325	-0.0034	-0.1568	-0.0210
SE_β	0.0154	0.0035	0.0049	0.0005	0.0246	0.0004
p -level	0.09	0.0003	2.17E-10	4.15E-11	8.4E-10	2.99E-06

the proximity of the Mauritanian coast explains the large differences observed in Chl-*a* levels and depth of the DCM between transects. The depth of the DCM ranged from 33 to 82 m, being deeper and with lower Chl-*a* concentration in the Western transect, as expected in oligotrophic open ocean waters (Antoine et al., 1996; Mignot et al., 2014; Iuculano et al., 2019). Conversely, the DCM was shallower and with higher Chl-*a* concentration in the Eastern transect, because of the impact of the offshore export of coastal plankton populations by the Cape Blanc filament (Gabric et al., 1993; Lovecchio et al., 2017; Santana-Falcón et al., 2020). The easternmost part of the Northern transect is dynamically very complex, with both cyclonic and anticyclonic eddies to the West and East of stn 5, respectively (Burgoa et al. in review). The presence of these eddies produces a convergence in stn 5 that generates the movement of the CVF to the North through this station (Fig. A10, Burgoa et al. in review). This mesoscale variability at stn 5 of the Northern transect is reflected in the higher concentration of Chl-*a*. The front interacts with the filament of Cape Blanc, which brought organic matter across the sampling point and generated a hydrographic anomaly around this station, which presents characteristics closer to those observed to the South of the front.

NH_4^+ and NO_2^- concentrations exhibited several maxima in the surroundings of the DCM. These maxima are associated with the enhanced microbial degradation processes of ammonification and nitrification at the DCM (Ward et al., 1982; Meeder et al., 2012). It has also been suggested that the primary nitrite maximum could be related to inefficient nitrate assimilative reduction by phytoplankton at the low light levels of the DCM (Lomas and Lipschultz, 2006). The mean concentrations of NH_4^+ ($0.06 \mu\text{mol kg}^{-1}$) and NO_2^- ($0.14 \mu\text{mol kg}^{-1}$) found throughout the epipelagic layer of the CVFZ were very low, as expected for oligotrophic waters (Twomey et al., 2007; Li et al., 2008). Higher levels of $0.8 \mu\text{mol kg}^{-1}$ for NH_4^+ and $1.5 \mu\text{mol kg}^{-1}$ for NO_2^- were found in the Eastern (coastal) transect, coinciding again with the offshore export of coastal waters by the filament of Cape Blanc. These levels are comparable with those found by Clark et al. (2016) in the mixed layer of a filament in the Mauritanian upwelling.

Higher DOC concentrations in the Eastern and Southern transect, i.e. near the coast and South of the CVF, were especially noticeable at the DCM. It is expected that DOC levels are higher when approaching to the coastal ocean (Gattuso et al., 1998; del Giorgio and Duarte, 2002) and can exceed $100 \mu\text{mol L}^{-1}$ in highly productive areas such as in EBUEs (Hill and Wheeler, 2002; Hansell et al., 2009). These high DOC levels can extend offshore due to the horizontal export by the filament of Cape Blanc (Gabric et al., 1993; Lovecchio et al., 2017; 2018). The C:N ratio of DOM was also higher in the Eastern and Southern transects. The mean C:N molar ratio of DOM at the CVFZ (11.7) is slightly higher than the value of 10 mol C mol N⁻¹ found by Church et al. (2002) and Hopkinson and Vallino (2005) in coastal and offshore regions of the North Atlantic and North Pacific central gyre. However, Letscher et al. (2013) found higher values around 16 mol C mol N⁻¹ in the upper 50 m of Eastern Subtropical North Atlantic, (10-40°N 10-50°W) and South Atlantic (10-40°S 10-50°W).

The highest concentrations of suspended POM were also observed near the coast. Van Camp et al. (1991) and Gabric et al. (1993) estimated

that the offshore currents associated with the giant filament could export around 50% of the POM produced in the coastal area around Cape Blanc (21°N). Furthermore, the differences of suspended POM concentration between the Northern and Southern transects could be explained by the passage of CVF across the Northern section around stn 5. The location of the POM maxima in the Eastern transects coincided with the DCM, while in the Western transect POM maxima were observed at the surface.

The C:N molar ratio of POM was very variable, being the lowest near the coast i.e. along the Eastern transect (7.6 ± 0.4), because of the intense production of fresh organic matter with a lower C:N ratio in coastal waters and subsequent offshore export by the filament of Cape Blanc. Furthermore, C:N molar ratios to the North of the CVF were higher than to the South and always higher than the canonical Redfield ratio of 6.6 (Redfield et al., 1963; Anderson, 1995). In fact, Martiny et al. (2013a, 2013b) showed large spatial variations in the C:N molar ratio of POM in the epipelagic ocean that differed substantially from the classical Redfield ratio, proposing a C:N:P ratio of 137:18:1 in warm, nutrient-rich upwelling zones. This C:N molar ratio of 7.6 is lower than the average value of 8.5, that we found down to the base of the DCM in the CVFZ.

4.2. Impact of water mass mixing in meso- and bathypelagic layers

We have applied a water mass analysis to obtain archetype DOC, DON, POC and PON concentrations for the 11 WTs identified during the FLUXES I cruise, for the entire cruise and for each transect. Our water mass analysis followed Pastor et al. (2012), except for the definition of the intermediate water AA and the deep waters. Concerning AA, its physical and chemical properties were taken in the Eastern Equatorial Atlantic, an appropriate source region because this modified Antarctic Intermediate water is transported from the equatorial region to the CVFZ by the Mauritanian current (Fig. 1). Conversely, Pastor et al. (2012) took the properties of the AA in the CVFZ. Concerning the deep waters, while Pastor et al. (2012) grouped all deep WTs under the generic denomination of North Atlantic Deep Water (NADW), we identified 3 WTs: UNEADW, LNEADW and LSW, with contrasting origin and silicate concentrations (Pérez-Rodríguez et al., 2001; Álvarez and Álvarez-Salgado, 2009). With regard to the central waters, ENACW and SACW differ in their thermohaline and chemical characteristics as a result of their different origin and history from their respective formation areas to the CVFZ (Stramma and Schott, 1999). ENACW is relatively more ventilated and, therefore, well oxygenated, which explains the contrasting O_2 concentrations in relation to their position with respect to the CVF. The coexistence of the different ENACW and SACW types was evident in the large-scale distribution of water properties in the FLUXES I hydrographic box, as well as at the mesoscale range, with the meandering of the CVF tracing the presence of low salinity central waters of southern origin in some stations of the Northern transect (Fig. 3, S1).

Archetype DOC concentration in ENACW ($48\text{--}52 \mu\text{mol L}^{-1}$) are lower than those found in the Eastern North Atlantic by Álvarez-Salgado et al. (2013) at 40°N and Lønborg and Álvarez-Salgado (2014) at 27°–42°N. This difference could be explained because ENACW at 40°N is closer to

the formation area and, therefore younger and DOC-richer, compared to the same water mass at 20°N. However, in intermediate and deep waters, the DOC concentrations were similar to those found by these authors. The archetype DOC concentrations explained a major fraction (63%) of the total DOC variability in the CVFZ, suggesting that mixing and basin-scale mineralisation processes occurring from the formation area to the CVFZ are the main drivers of the DOC variability.

POC concentrations in the mesopelagic layer of the CVFZ are low, with an average value of $2.0 \pm 0.5 \mu\text{mol L}^{-1}$. [Alonso-González et al. \(2009\)](#) reported higher concentrations of suspended POC in the mesopelagic layer of the Canary current from 20° to 29°N, ($3\text{--}8 \mu\text{mol L}^{-1}$). This discrepancy is probably due the location of both sampling areas, in the core of the Canary current in [Alonso-González et al. \(2009\)](#) and further South in our study. The FLUXES I box is affected by the giant filament of Cape Blanc, where higher contribution of sinking particulate material should be expected ([Fischer et al., 2009](#)). Moreover, these high sinking rates are favoured by ballasting effect of atmospheric dust inputs in the area ([Bory and Newton, 2000](#); [Fischer and Karakaş, 2009](#)). In contrast, POC concentrations in bathypelagic layers of the CVFZ are consistent with what is described by [Alonso-González et al. \(2009\)](#).

The C:N ratios of suspended POM and DOM are higher than the canonical Redfield ratio and increase with depth indicating PON and DON depletion with respect to POC and DOC due to preferential remineralisation of nitrogen-rich compounds ([Álvarez-Salgado et al., 2013](#); [Hopkinson and Vallino, 2005](#); [Letscher et al., 2013](#); [Lønborg and Álvarez-Salgado, 2014](#)). Previous research in the study area also found elevated, and sometimes depth dependent, C:N ratios of sinking POM, higher than the Redfield ratio ([Fischer et al., 2009, 2019](#); [Nowald et al., 2015](#)), but always lower than the C:N ratios of suspended POM found in this work. The same trend of higher C:N ratios of suspended compared to sinking POM was observed by [Schneider et al. \(2003\)](#) in the subtropical gyre of the North Pacific. We hypothesise that the larger residence time of small size suspended particles in the water column exacerbates the impact of preferential nitrogen mineralisation, explaining the higher C:N molar ratios observed.

4.3. Remineralisation of organic matter in the CVFZ

The DOC to AOU ratio obtained from the multiple linear correlation of DOC with the WT proportions and AOU suggests that DOC mineralisation explains only $3.6 \pm 2.1\%$ of the inorganic carbon production in the meso- and bathypelagic waters of the CVFZ. This surprisingly low contribution is well below the 10–20% found by [Aristegui et al. \(2002\)](#) for the world ocean, the 26.5% to the south of the Canary Islands obtained by [Aristegui et al. \(2003\)](#), the 9–19% obtained by [Carlson et al. \(2010\)](#) in the deep North Atlantic, the $28 \pm 3\%$ found by [Álvarez-Salgado et al. \(2013\)](#) in the subpolar North Atlantic or the $26 \pm 3\%$ by [Lønborg and Álvarez-Salgado \(2014\)](#) in the Eastern North Atlantic from 27° to 42°N. The low contribution of DOC mineralisation in the CVFZ could be explained by 1) the fact that most of the water masses in this region are far away from their formation areas, where the proportion of labile DOC is higher ([Hansell et al., 2012](#); [Hansell and Carlson, 2013](#)). In fact, we have shown that archetype DOC concentrations in the ENACW of the CVFZ are significantly lower than in areas closer to the formation area of these WTs ([Álvarez-Salgado et al., 2013](#); [Lønborg and Álvarez-Salgado, 2014](#)). And 2) the massive flux of biodegradable sinking POM produced in the coast, which is subjected to lateral export by the giant filament of Cape Blanc and sinks as the filament is moving away from the continental shelf, ballasted lithogenic materials ([Iversen et al., 2010](#); [Álvarez-Salgado and Aristegui, 2015](#); [Lovecchio et al., 2018](#)). As recently observed by [Lopez et al. \(2020\)](#) in the Eastern North Pacific, the DOC distribution in the CVFZ is apparently affected by the dissolution of these fast sinking particles, which generate columns of DOC to the bottom.

In the case of POC, it represents $4.5 \pm 0.7\%$ of the inorganic carbon production of the meso- and bathypelagic waters of the CVFZ. This

contribution is far below the estimated by [Alonso-González et al. \(2009\)](#). They estimated that suspended POC transported laterally could account for 28% to 59% of the total mesopelagic respiration in the Canary region. The low contribution in the CVFZ suggests that the respiratory organic carbon demand in the dark ocean is mainly supported by sinking POM. In contrast, suspended POC may be exported rapidly offshore in this area of strong offshore zonal flow, leaving a minor imprint in the biogeochemistry of the study area. This situation contrasts with the Northern location in [Alonso-González et al. \(2009\)](#), where the dominant mean flow is Southward and along-shore with the Canary Current, and the zonal transport is localized in mesoscale filaments and eddies ([Lovecchio et al. 2017](#), [Álvarez-Salgado et al. 2007](#)).

The DON to NO_3^- coefficient indicates that DON supported 16% of NO_3^- produced following organic matter mineralisation. [Álvarez-Salgado et al. \(2006\)](#) found that in NW Iberian upwelling (42°N), the contribution of DOM to the mineralisation of organic nitrogen was 30%, while in deep open oceans waters of the North Pacific, the contribution of DON to nitrate production was from 10% to 25% ([Jackson and Williams, 1985](#); [Maita and Yanada, 1990](#)). Mineralized PON represented only the $2.1 \pm 0.4\%$ of NO_3^- . As for the case of POC with the oxygen utilisation, the maximum contribution of mineralised PON to NO_3^- was $3.9 \pm 1.1\%$ in the Northern transect. The higher contributions of mineralised DON/PON to NO_3^- than POC/PON to oxygen utilisation support the idea that the preferential N remineralisation explains the higher Redfield POM/DOM C:N ratios in the CVFZ.

5. Conclusions

The distribution of DOM and suspended POM in the CVFZ is dictated by 1) the position of the front, which separates surface and central waters of contrasting North and South Atlantic origin, and 2) the intersection with meanders of the frontal system associated with mesoscale structures and their interaction with the giant upwelling filament of Cape Blanc. In the intermediate and deep waters, which are very distant from their respective source regions, the distributions are dictated by water mass mixing and remineralization from the source regions to the CVFZ. DOM and suspended POM mineralisation in the study hydrographic box represent only 8.1% of the carbon mineralisation and 17.8% of the nitrogen mineralisation, suggesting that 1) the local carbon demand is mainly supported by sinking POM and 2) N-containing organic compounds are mineralised faster than C-containing organic compounds.

The results obtained in this field study confirm a latitudinal gradient in the importance of the vertical vs. horizontal flows of organic matter in the Canary Current EBUE, and suggest that the offshore regions of EBUEs act as a transitional zone that modifies the stoichiometry of the organic matter exported to the gyres, making it poor in nitrogen.

CRedit authorship contribution statement

S. Valiente: Methodology, Investigation, Formal analysis, Writing – original draft, Visualization. **B. Fernández-Castro:** Methodology, Formal analysis, Investigation, Writing – review & editing, Visualization. **R. Campanero:** Investigation, Writing – review & editing. **A. Marrero-Díaz:** Data curation, Investigation, Writing – review & editing. **A. Rodríguez-Santana:** Methodology, Investigation, Writing – review & editing. **M.D. Gelado-Cabellero:** Methodology, Investigation, Writing – review & editing. **M. Nieto-Cid:** Conceptualization, Methodology, Investigation, Writing – review & editing. **A. Delgado-Huertas:** Conceptualization, Writing – review & editing, Supervision, Funding acquisition. **J. Aristegui:** Conceptualization, Methodology, Investigation, Writing – review & editing, Project administration, Funding acquisition. **X.A. Álvarez-Salgado:** Conceptualization, Methodology, Formal analysis, Investigation, Writing – original draft, Writing – review & editing, Visualization, Supervision, Project administration, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

We would like to thank the captain and crew of R/V Sarmiento de Gamboa and the personnel of the CSIC Unidad de Tecnología Marina (UTM), for their help during the FLUXES I cruise, as well as to M.J. Pazo for DOM and POM analysis and V. Vieitez for nutrient and POM analyses. This work was supported by Spanish National Science Plan research grants FLUXES (CTM2015-69392-C3) and e-IMPACT (PID2019-109084RB-C2). S.V.R. and R.C.N. were supported by PhD fellowships from the Spanish Ministry of Science and Innovation (BES-2016-079216 and BES-2016-076462); B.F.C. was supported by a Juan de La Cierva Formación fellowship (FJCI-641-2015-25712) and by the European Union's Horizon 2020 research and innovation program under the Marie Skłodowska-Curie grant agreement No. 834330 (SO-CUP). M.N.-C. was partially supported by the project FERMIO (MINECO, CTM2014-57334-JIN), co-financed with FEDER funds. J.A. was partly supported by the project SUMMER (AMD-817806-5) from the European Union's Horizon 2020 research and innovation program.

Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.pocean.2021.102727>.

References

- Alonso-González, I.J., Arístegui, J., Vilas, J.C., Hernández-Guerra, A., 2009. Lateral POC transport and consumption in surface and deep waters of the Canary Current region: A box model study. *GB2007 Global Biogeochem. Cycles* 23. <https://doi.org/10.1029/2008GB003185>.
- Alpers, W., Brandt, P., Lazar, A., Dagorne, D., Sow, B., Faye, S., Hansen, M.W., Rubino, A., Poulain, P.M., Brehmer, P., 2013. A small-scale oceanic eddy off the coast of West Africa studied by multi-sensor satellite and surface drifter data. *Remote Sens. Environ.* 129, 132–143. <https://doi.org/10.1016/j.rse.2012.10.032>.
- Álvarez-Salgado, X.A., Arístegui, J., Barton, E.D., Hansell, D.A., 2007. Contribution of upwelling filaments to offshore carbon export in the subtropical Northeast Atlantic Ocean. *Limnol. Oceanogr.* 52, 1287–1292. <https://doi.org/10.4319/lo.2007.52.3.1287>.
- Álvarez-Salgado, X.A., Nieto-Cid, M., Gago, J., Brea, S., Castro, C.G., Doval, M.D., Pérez, F.F., 2006. Stoichiometry of the degradation of dissolved and particulate biogenic organic matter in the NW Iberian upwelling. *J. Geophys. Res.* 111, C07017. <https://doi.org/10.1029/2004JC002473>.
- Álvarez-Salgado, X., Arístegui, J., 2015. Organic matter dynamics in the Canary Current. In: Vald, L., Déniz-González, I. (Eds.), *Oceanogr. Biol. Featur. Canar. Curr. Large Mar. Ecosyst. IOC-UNESCO, Paris*, pp. 151–160 (IOC Tech. Ser. No. 115).
- Álvarez-Salgado, X.A., Nieto-Cid, M., Álvarez, M., Pérez, F.F., Morin, P., Mercier, H., 2013. New insights on the mineralization of dissolved organic matter in central, intermediate, and deep water masses of the northeast North Atlantic. *Limnol. Oceanogr.* 58 (2), 681–696. <https://doi.org/10.4319/lo.2013.58.2.0681>.
- Álvarez, M., Álvarez-Salgado, X., 2009. Chemical tracer transport in the eastern boundary current system of the North Atlantic. *Ciencias Mar.* 35, 123–139. <https://doi.org/10.7773/cm.v35i2.1438>.
- Álvarez, M., Brea, S., Mercier, H., Álvarez-Salgado, X., 2014. Mineralization of biogenic materials in the water masses of the South Atlantic Ocean. I: Assessment and results of an optimum multiparameter analysis. *Prog. Oceanogr.* 123, 1–23. <https://doi.org/10.1016/j.pocean.2013.12.007>.
- Anderson, L.A., 1995. On the hydrogen and oxygen content of marine phytoplankton. *Deep. Res. Part I* 42 (9), 1675–1680. [https://doi.org/10.1016/0967-0637\(95\)00072-E](https://doi.org/10.1016/0967-0637(95)00072-E).
- Antoine, D., André, J.-M., Morel, A., 1996. Oceanic primary production: 2. Estimation at global scale from satellite (Coastal Zone Color Scanner) chlorophyll. *Global Biogeochem. Cycles* 10 (1), 57–69. <https://doi.org/10.1029/95GB02832>.
- Arístegui, J., Barton, E.D., Montero, M.F., García-Muñoz, M., Escáñez, J., 2003. Organic carbon distribution and water column respiration in the NW Africa-Canaries Coastal Transition Zone. *Aquat. Microb. Ecol.* 33, 289–301.
- Arístegui, J., Barton, E.D., Álvarez-Salgado, X.A., Santos, A.M.P., Figueiras, F.G., Kifani, S., Hernández-León, S., Mason, E., Machú, E., Demarcq, H., 2009. Sub-regional ecosystem variability in the Canary Current upwelling. *Prog. Oceanogr.* 83 (1–4), 33–48. <https://doi.org/10.1016/j.pocean.2009.07.031>.
- Arístegui, J., Duarte, C., Agustí, S., Doval, M., Álvarez-Salgado, X., Hansell, D.A., 2002. Dissolved organic carbon support of respiration in the dark ocean. *Science* 298, 967.
- Arístegui, J., Montero, M., Hernández-Hernández, N., Alonso-González, I., Baltar, F., Calleja, M.L., Duarte, C.M., 2020. Variability in water-column respiration and its dependence on organic carbon sources in the Canary Current upwelling region. *Front. Earth Sci.* 8, 349. <https://doi.org/10.3389/feart.2020.00349>.
- Barton, E.D., Arístegui, J., Tett, P., Cantón, M., García-Braun, J., Hernández-León, S., Nykjaer, L., Almeida, C., Almunia, J., Ballesteros, S., Basterretxea, G., Escáñez, J., García-Weill, L., Hernández-Guerra, A., López-Laatzén, F., Molina, R., Montero, M. F., Navarro-Pérez, E., Rodríguez, J.M., van Lenning, K., Vélez, H., Wild, K., 1998. The transition zone of the Canary Current upwelling region. *Prog. Oceanogr.* 41 (4), 455–504.
- Bory, A.-M., Newton, P.P., 2000. Transport of airborne lithogenic material down through the water column in two contrasting regions of the eastern subtropical North Atlantic Ocean. *Global Biogeochem. Cycles* 14 (1), 297–315. <https://doi.org/10.1029/1999GB900098>.
- Boyd, P.W., Claustre, H., Levy, M., Siegel, D.A., Weber, T., 2019. Multi-faceted particle pumps drive carbon sequestration in the ocean. *Nature* 568 (7752), 327–335.
- Broecker, W.S., 1974. NO a conservative water-mass tracer. *Earth Planet. Sci. Lett.* 23 (1), 100–107.
- Burgoa, N., Machín, F., Marrero-Díaz, S., Rodríguez-Santana, A., Martínez-Marrero, A., Arístegui, J., Duarte, C., 2020. Mass, nutrients and dissolved organic carbon (DOC) lateral transports off northwest Africa during fall 2002 and spring 2003 Item Type Article. *Ocean Sci.* 16, 483–511. <https://doi.org/10.5194/os-16-483-2020>.
- Cardoso, C., Caldeira, R.M.A., Relvas, P., Stegner, A., 2020. Islands as eddy transformation and generation hotspots: Cabo Verde case study. *Prog. Oceanogr.* 184, 102271. <https://doi.org/10.1016/j.pocean.2020.102271>.
- Carlson, C.A., Hansell, D.A., Nelson, N.B., Siegel, D.A., Smethie, W.M., Khaliwala, S., Meyers, M.M., Halewood, E., 2010. Dissolved organic carbon export and subsequent remineralization in the mesopelagic and bathypelagic realms of the North Atlantic basin. *Deep Sea Res. Part II* 57 (16), 1433–1445.
- Church, M.J., Ducklow, H.W., Karl, D.M., 2002. Multiyear increases in dissolved organic matter inventories at Station ALOHA in the North Pacific Subtropical Gyre. *Limnol. Oceanogr.* 47 (1), 1–10. <https://doi.org/10.4319/lo.2002.47.1.0001>.
- Clark, D.R., Widdicombe, C.E., Rees, A.P., Woodward, E., 2016. The significance of nitrogen regeneration for new production within a filament of the Mauritanian upwelling system. *Biogeosciences* 13, 2873–2888. <https://doi.org/10.5194/bg-13-2873-2016>.
- del Giorgio, P.A., Duarte, C.M., 2002. Respiration in the open ocean. *Nature* 420 (6914), 379–384.
- Fischer, G., Karakas, G., 2009. Sinking rates and ballast composition of particles in the Atlantic Ocean: implications for the organic carbon fluxes to the deep ocean. *Biogeosciences* 6, 85–102. <https://doi.org/10.5194/bg-6-85-2009>.
- Fischer, G., Reuter, C., Karakas, G., Nowald, N., Wefer, G., 2009. Offshore advection of particles within the Cape Blanc filament, Mauritania: Results from observational and modelling studies. *Prog. Oceanogr.* 83 (1–4), 322–330.
- Fischer, G., Romero, O., Toby, E., Iversen, M., Donner, B., Mollenhauer, G., Nowald, N., Ruhland, G., Klann, M., Hamady, B., Wefer, G., 2019. Changes in the dust-influenced biological carbon pump in the canary current system: implications from a coastal and an offshore sediment trap record off cape Blanc, Mauritania. *Global Biogeochem. Cycles* 33 (8), 1100–1128. <https://doi.org/10.1029/2019GB006194>.
- Gabric, A.J., Garcia, L., Van Camp, L., Nykjaer, L., Eiffel, W., Schrimpf, W., 1993. Offshore export of shelf production in the Cape Blanc (Mauritania) giant filament as derived from coastal zone color scanner imagery. *J. Geophys. Res.* 98 (C3), 4697–4712. <https://doi.org/10.1029/92JC01714>.
- Gattuso, J.-P., Frankignoulle, M., Wollast, R., 1998. Carbon and Carbonate metabolism in coastal aquatic ecosystems. *Annu. Rev. Ecol. Syst.* 29 (1), 405–434. <https://doi.org/10.1146/annurev.ecolsys.29.1.405>.
- Hansell, D.A., Carlson, C.A., 2013. Localized refractory dissolved organic carbon sinks in the deep ocean. *Global Biogeochem. Cycles* 27 (3), 705–710. <https://doi.org/10.1002/gbc.20067>.
- Hansell, D.A., Carlson, C.A., Schlitzer, R., 2012. Net removal of major marine dissolved organic carbon fractions in the subsurface ocean. *GB1016 Global Biogeochem. Cycles* 26. <https://doi.org/10.1029/2011GB004069>.
- Hansell, D., Carlson, C., Repeta, D., Schlitzer, R., 2009. Dissolved organic matter in the ocean: A controversy stimulates new insights. *Oceanography* 22 (4), 202–211.
- Hansen, H., Koroleff, F., 1999. *Methods of seawater analysis*, 3rd. Wiley, Weinheim, pp. 149–228.
- Helmke, P., Romero, O., Fischer, G., 2005. Northwest African upwelling and its effect on offshore organic carbon export to the deep sea. *GB4015 Global Biogeochem. Cycles* 19. <https://doi.org/10.1029/2004GB002665>.
- Hill, J.K., Wheeler, P.A., 2002. Organic carbon and nitrogen in the northern California current system: comparison of offshore, river plume, and coastally upwelled waters. *Prog. Oceanogr.* 53 (2–4), 369–387.
- Holm-Hansen, O., Lorenzen, C.J., Holmes, R.W., Strickland, J.D.H., 1965. Fluorometric determination of chlorophyll. *ICES J. Mar. Sci.* 30 (1), 3–15. <https://doi.org/10.1093/icesjms/30.1.3>.
- Hopkinson, C.S., Vallino, J.J., 2005. Efficient export of carbon to the deep ocean through dissolved organic matter. *Nature* 433 (7022), 142–145.
- Iuculano, F., Álvarez-Salgado, X., Otero, J., Catalá, T., Sobrino, C., Duarte, C., Agustí, S., 2019. Patterns and drivers of UV absorbing chromophoric dissolved organic matter in the euphotic layer of the open ocean. *Front. Mar. Sci.* 6, 320.
- Iversen, M.H., Nowald, N., Ploug, H., Jackson, G.A., Fischer, G., 2010. High resolution profiles of vertical particulate organic matter export off Cape Blanc, Mauritania: Degradation processes and ballasting effects. *Deep. Res. Part I* 57 (6), 771–784.

- Jackson, G.A., Williams, P.M., 1985. Importance of dissolved organic nitrogen and phosphorus to biological nutrient cycling. *Deep Sea Res. Part A Oceanogr. Res. Pap.* 32 (2), 223–235. [https://doi.org/10.1016/0198-0149\(85\)90030-5](https://doi.org/10.1016/0198-0149(85)90030-5).
- Karakas, G., Nowald, N., Blaas, M., Marchesio, P., Frickenhaus, S., Schlitzer, R., 2006. High-resolution modeling of sediment erosion and particle transport across the northwest African shelf. *J. Geophys. Res.* 111, C06025. <https://doi.org/10.1029/2005JC003296>.
- Karstensen, J., Tomczak, M., 1998. Age determination of mixed water masses using CFC and oxygen data. *J. Geophys. Res.* 103 (C9), 18599–18609. <https://doi.org/10.1029/98JC00889>.
- Kérouel, R., Aminot, A., 1997. Fluorometric determination of ammonia in sea and estuarine waters by direct segmented flow analysis. *Mar. Chem.* 57 (3–4), 265–275. [https://doi.org/10.1016/S0304-4203\(97\)00040-6](https://doi.org/10.1016/S0304-4203(97)00040-6).
- Langdon, C., 2010. Determination of dissolved oxygen in seawater by winkler titration using the amperometric technique. In: Hood, E.M., Sloyan, B.M., Sabine, C. (Eds.), *Go-sh. Repeat Hydrogr. Man. A Collect. Expert Reports Guidel. IOCCP Rep. no. 14*, ICPO Publ. Ser. no. 134.
- Letscher, R.T., Hansell, D.A., Carlson, C.A., Lumpkin, R., Knapp, A.N., 2013. Dissolved organic nitrogen in the global surface ocean: Distribution and fate. *Global Biogeochem. Cycles* 27 (1), 141–153. <https://doi.org/10.1029/2012GB004449>.
- Li, Q.P., Hansell, D.A., Zhang, J.-Z., 2008. Underway monitoring of nanomolar nitrate plus nitrite and phosphate in oligotrophic seawater. *Limnol. Oceanogr. Methods* 6 (7), 319–326. <https://doi.org/10.4319/lom.2008.6.319>.
- Lomas, M.W., Lipschultz, F., 2006. Forming the primary nitrite maximum: Nitrifiers or phytoplankton? *Limnol. Oceanogr.* 51 (5), 2453–2467. <https://doi.org/10.4319/lo.2006.51.5.2453>.
- Lønborg, C., Álvarez-Salgado, 2014. Tracing dissolved organic matter cycling in the eastern boundary of the temperate North Atlantic using absorption and fluorescence spectroscopy. *Deep. Res. I* 85, 35–46.
- Lopez, C.N., Robert, M., Galbraith, M., Bercovici, S.K., Orellana, M.V., Hansell, D.A., 2020. High temporal variability of total organic carbon in the deep Northeastern Pacific. *Front. Earth Sci.* 8, 80. <https://doi.org/10.3389/feart.2020.00080>.
- Lovecchio, Elisa, Gruber, Nicolas, Münnich, Matthias, 2018. Mesoscale contribution to the long-range offshore transport of organic carbon from the Canary Upwelling System to the open North Atlantic. *Biogeosciences* 15 (16), 5061–5091. <https://doi.org/10.5194/bg-15-5061-2018>. <https://doi.org/10.5194/bg-15-5061-2018-supplement>.
- Lovecchio, E., Gruber, N., Münnich, M., Lachkar, Z., 2017. On the long-range offshore transport of organic carbon from the Canary Upwelling System to the open North Atlantic. *Biogeosciences* 14, 3337–3369. <https://doi.org/10.3929/ethz-b-000190480>.
- Maita, Yoshiaki, Yanada, Mitsuru, 1990. Vertical distribution of total dissolved nitrogen and dissolved organic nitrogen in seawater. *Geochim. J.* 24 (4), 245–254. <https://doi.org/10.2343/geochimj.24.245>.
- Martínez-Marrero, A., Rodríguez-Santana, A., Hernández-Guerra, A., Fraile-Nuez, E., López-Laatzén, F., Vélez-Belchí, P., Parrilla, G., 2008. Distribution of water masses and diapycnal mixing in the Cape Verde Frontal Zone. *107609 Geophys. Res. Lett.* 35. <https://doi.org/10.1029/2008GL033229>.
- Martiny, A., Vrugt, J., Primeau, F., Lomas, M., 2013a. Regional variation in the particulate organic carbon to nitrogen ratio in the surface ocean. *Global Biogeochem. Cycles* 27, 723–731. <https://doi.org/10.1002/gbc.20061>.
- Martiny, A.C., Pham, C.T.A., Primeau, F.W., Vrugt, J.A., Moore, J.K., Levin, S.A., Lomas, M.W., 2013b. Strong latitudinal patterns in the elemental ratios of marine plankton and organic matter. *Nat. Geosci.* 6, 279–283. <https://doi.org/10.1038/ngeo1757>.
- Meeder, E., Mackey, K., Paytan, A., Shaked, Y., Iluz, D., Stambler, N., Rivlin, T., Post, A., Lazar, B., 2012. Nitrite dynamics in the open ocean—clues from seasonal and diurnal variations. *Mar. Ecol. Prog. Ser.* 453, 11–26. <https://doi.org/10.3354/meps09525>.
- Meunier, T., Barton, E., Barreiro, B., Torres, R., 2012. Upwelling filaments off Cape Blanc: Interaction of the NW African upwelling current and the Cape Verde frontal zone eddy field? *C08031 J. Geophys. Res.* 117. <https://doi.org/10.1029/2012JC007905>.
- Mignot, Alexandre, Claustre, Hervé, Uitz, Julia, Poteau, Antoine, D'Ortenzio, Fabrizio, Xing, Xiaogang, 2014. Understanding the seasonal dynamics of phytoplankton biomass and the deep chlorophyll maximum in oligotrophic environments: A Bio-Argo float investigation. *Global Biogeochem. Cycles* 28 (8), 856–876. <https://doi.org/10.1002/2013GB004781>.
- Nowald, N., Iversen, M., Fischer, G., Ratmeyer, V., Wefer, G., 2015. Time series of in-situ particle properties and sediment trap fluxes in the coastal upwelling filament off Cape Blanc, Mauritania. *Prog. Oceanogr.* 137, 1–11. <https://doi.org/10.1016/j.pocean.2014.12.015>.
- Nowald, N., Karakas, G., Ratmeyer, V., Fischer, G., Schlitzer, R., Davenport, R.A., Wefer, G., 2006. Distribution and transport processes of marine particulate matter off Cape Blanc (NW-Africa): results from vertical camera profiles. *Ocean Sci. Discuss* 3, 903–938.
- Ohde, Thomas, Fiedler, Björn, Körtzinger, Arne, 2015. Spatio-temporal distribution and transport of particulate matter in the eastern tropical North Atlantic observed by Argo floats. *Deep. Res. Part I* 102, 26–42.
- Olivar, M., Sabatés, A., Pastor, M., Pelegrí, J., 2016. Water masses and mesoscale control on latitudinal and cross-shelf variations in larval fish assemblages off NW Africa. *Deep Res. Part I Oceanogr. Res. Pap.* 117, 120–137. <https://doi.org/10.1016/j.dsr.2016.10.003>.
- Pastor, M., Peña-Izquierdo, J., Pelegrí, J., Marrero-Díaz, A., 2012. Meridional changes in water mass distributions off NW Africa during November 2007/2008. *Ciencias Mar.* 38, 223–244. <https://doi.org/10.7773/cm.v38i1B.1831>.
- Pelegrí, J.L., Peña-Izquierdo, J., 2015. Easter boundary currents off North West Africa. *Oceanogr. Biol. Featur. Canar. Curr. Large Mar. Ecosyst.* 3, 81–92. *IOC Tech. Ser.* 115 81–92 IOC Technical Series.
- Pérez-Rodríguez, P., Pelegrí, J., Marrero-Díaz, A., 2001. Dynamical characteristics of the Cape Verde frontal zone. *Mar. Sci.* 65, 241–250.
- Redfield, A.C., Ketchum, B., Richards, F., 1963. The influence of organisms on the composition of seawater. Wiley-Interscience, New York, pp. 26–77. Richards, F. A.
- Repeta, D.J., 2015. Chemical characterization and cycling of dissolved organic matter. In: *Biogeochemistry of Marine Dissolved Organic Matter*, second ed. Elsevier Inc., pp. 21–63. <https://doi.org/10.1016/B978-0-12-405940-5.00002-9>.
- Sangrà, Pablo, Pascual, Ananda, Rodríguez-Santana, Ángel, Machín, Francisco, Mason, Evan, McWilliams, James C., Pelegrí, Josep L., Dong, Changming, Rubio, Anna, Aristegui, Javier, Marrero-Díaz, Ángeles, Hernández-Guerra, Alonso, Martínez-Marrero, Antonio, Auladell, Maricel, 2009. The Canary Eddy Corridor: A major pathway for long-lived eddies in the subtropical North Atlantic. *Deep. Res. Part I Oceanogr. Res. Pap.* 56 (12), 2100–2114. <https://doi.org/10.1016/j.dsr.2009.08.008>.
- Santana-Falcón, Yera, Mason, Evan, Aristegui, Javier, 2020. Offshore transport of organic carbon by upwelling filaments in the Canary Current System. *Prog. Oceanogr.* 186, 102322. <https://doi.org/10.1016/j.pocean.2020.102322>.
- Schlitzer, R., 2017. Ocean Data View. Ocean Data View.
- Schneider, Birgit, Schlitzer, Reiner, Fischer, Gerhard, Nöthig, Eva-Maria, 2003. Depth-dependent elemental compositions of particulate organic matter (POM) in the ocean. *Global Biogeochem. Cycles* 17. <https://doi.org/10.1029/2002GB001871>.
- Stramma, Lothar, Schott, Friedrich, 1999. The mean flow field of the tropical Atlantic Ocean. *Deep. Res. II* 46 (1–2), 279–303.
- Tiedemann, Maik, Fock, Heino O., Döring, Julian, Badji, Luc Bonaventure, Möllmann, Christian, 2018. Water masses and oceanic eddy regulation of larval fish assemblages along the Cape Verde Frontal Zone. *J. Mar. Syst.* 183, 42–55.
- Tomczak, M., 1999. Some historical, theoretical and applied aspects of quantitative water mass analysis. *J. Mar. Res.* 57, 275–303. <https://doi.org/10.1357/002224099321618227>.
- Twomey, L.J., Waite, A.M., Pez, V., Pattiaratchi, C.B., 2007. Variability in nitrogen uptake and fixation in the oligotrophic waters off the south west coast of Australia. *Deep Res. Part II Top. Stud. Oceanogr.* 54 (8–10), 925–942. <https://doi.org/10.1016/j.dsr2.2006.10.001>.
- UNESCO, 1986. Progression Oceanographic Tables and Standards 1983–1986. Work and Recommendations of the UNESCO/SCOR/ICES/IAPSO Joint Panel. UNESCO Technical Papers in Marine Science 50.
- Van Camp, L., Nykjaer, L., Mittelstaedt, E., Schlittenhardt, P., 1991. Upwelling and boundary circulation off Northwest Africa as depicted by infrared and visible satellite observations. *Prog. Oceanogr.* 26 (4), 357–402.
- Verdugo, Pedro, Alldredge, Alice L., Azam, Farooq, Kirchman, David L., Passow, Uta, Santschi, Peter H., 2004. The oceanic gel phase: a bridge in the DOM-POM continuum. *Mar. Chem.* 92 (1–4), 67–85. <https://doi.org/10.1016/j.marchem.2004.06.017>.
- Ward, B.B., Olson, R.J., Perry, M.J., 1982. Microbial nitrification rates in the primary nitrite maximum off southern California. *Deep Sea Res. Part A Oceanogr. Res. Pap.* 29 (2), 247–255. [https://doi.org/10.1016/0198-0149\(82\)90112-1](https://doi.org/10.1016/0198-0149(82)90112-1).
- World Ocean Atlas, 2013. <https://www.nodc.noaa.gov/OC5/woa13/>.
- Zenk, W., Klein, B., Schroder, M., 1991. Cape verde frontal zone. *Deep Sea Res. Part A* 38, S505–S530. [https://doi.org/10.1016/S0198-0149\(12\)80022-7](https://doi.org/10.1016/S0198-0149(12)80022-7).